

COMP20008 Elements of Data Processing



Lecture 9: Unsupervised: Clustering and PCA

Semester 2 2026

Subject Coordinator: **Dr Ekaterina Vylomova**

Co-lecturer: **Dr Hanzhuo (Vivian) Ma**



Reminders

- By **this FRIDAY 5PM:**
 - Assignment 1 code submission
 - Finalise your group for A2 and register it!

Announcements

- **More consultation sessions:**

- Student Hub (Melbourne Connect; Level 3):

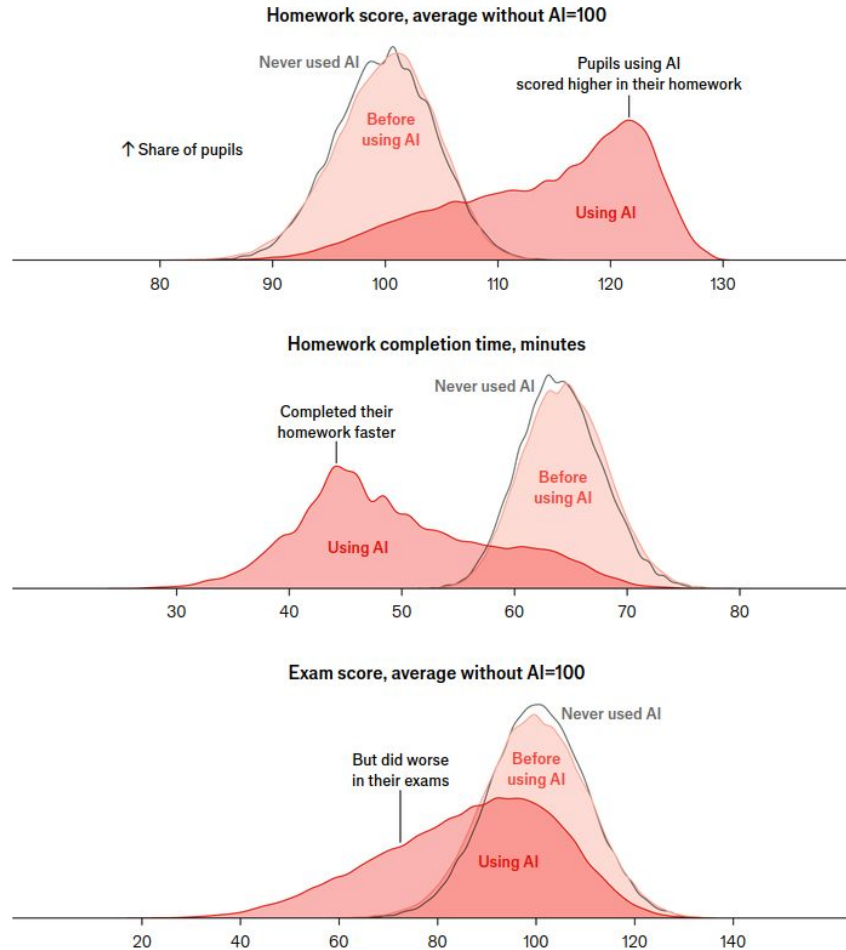
- Monday 1–3pm

- Friday 1–3pm

- Hub: <https://cis.unimelb.edu.au/study/cis-student-hub>

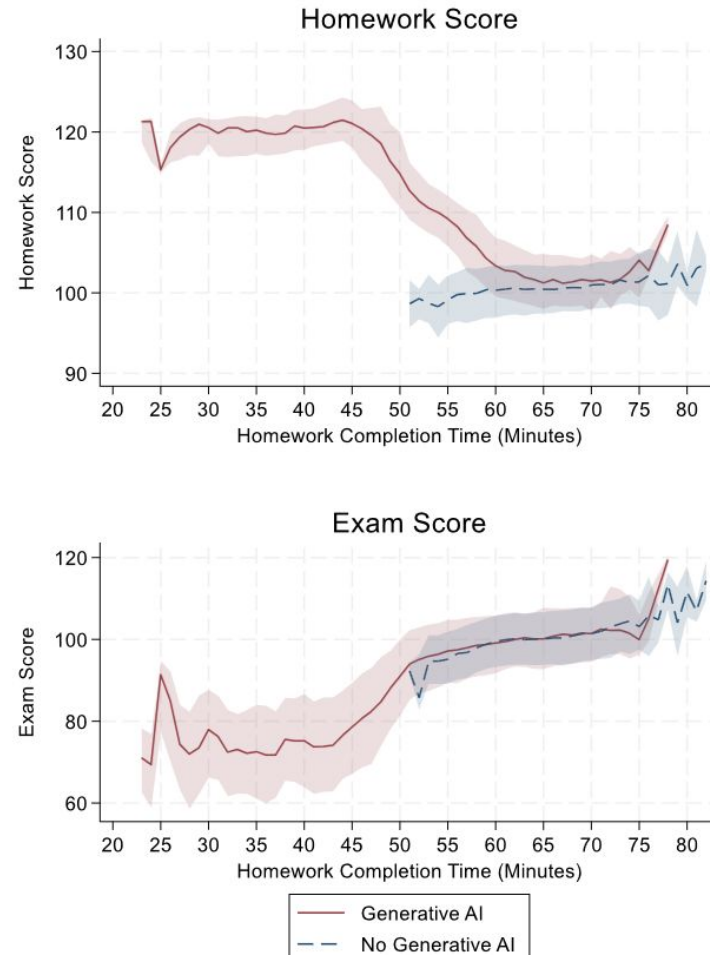
- After Monday Lecture, i.e. 6pm, same theatre

AI & Learning



<https://archive.is/wleng>

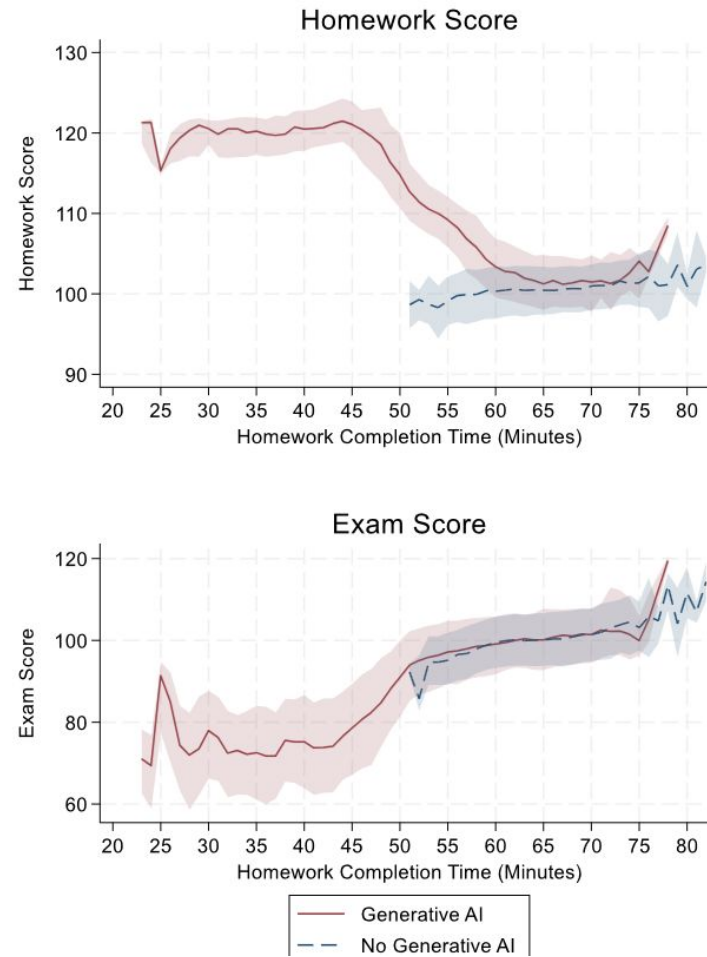
AI & Learning



https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6868618

AI & Learning

Use AI Wisely!



https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6868618

Recap: distributions

A1: AirBnB property listings

#	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC
1	id	name	description	host_id	host_is_nsfw	host_listing_id	neighbourhood	latitude	longitude	property_type	room_type	accommodates	bathroom	bedrooms	beds	amenities	price	minimum_stay	maximum_stay	availability	has_availability	number_of_reviews	first_review	last_review	review_score	review_score	review_score	review_score	reviews_per_month
2	10803	Room in D A large air	38901 t	f		1	Moreland	-37.7661	144.9795	Private room	Private room	2	1 shared bath		1	["Self check in", "Dishes and silverware", "Heating", "Hair dryer"]	\$60.74	5	28	158 t	214	#####	4/5/2026	4.57	4.07	4.66	1.31		
3	12936	St Kilda 18 RIGHT IN T	50121 t	f		10	Port Phillip	-37.86	144.9766	Entire rental unit	Entire rental unit	2	1 bath		1	["Self check in", "Dishes and silverware", "Heating", "Hair dryer"]	\$165.00	2	1125	278 t	280	#####	14/5/2026	4.88	4.94	4.86	1.48		
4	41836	CLOSE TO Easy to tra	182833 t	f		3	Darebin	-37.6976	145.0007	Private room	Private room	2	1 shared bath			["Heating", "Hair dryer"]	\$215.00	1	14	169 t	67	9/9/2010	17/5/2026	4.77	4.83	4.71	0.35		
5	43429	Tranquil Ja No service	189684 t	t		4	Monash	-37.8998	145.1158	Entire rental unit	Entire rental unit	2	1 bath		1	["Self check in", "Long term stays allc"]	\$165.00	4	21	0 t	180	9/9/2010	8/1/2025	4.88	4.92	4.81	0.94		
6	44699	15 yearsH Unwaverir	189245 t	f		3	Port Phillip	-37.8316	144.9706	Private room	Private room	2	1 shared bath		2	["Bay view", "Long term stays allc"]	\$215.00	1	14	169 t	67	9/9/2010	17/5/2026	4.77	4.83	4.71	0.35		
7	47100	Cosy, cute Our home	212071 t	f		2	Yarra	-37.8184	145.005	Entire rental unit	Entire rental unit	2	1 bath		1	["Long term stays allc"]	\$294.10	2	1125	208 t	537	22/2/2011	7/6/2026	4.91	4.75	4.79	2.88		
8	51592	Central Cit Stay in a st	190879 t	t		2	Melbourne	-37.8127	144.9631	Entire loft	Entire rental unit	2	1.5 baths		1	["Self check in", "Long term stays allc"]	\$294.10	2	1125	208 t	537	22/2/2011	7/6/2026	4.91	4.75	4.79	2.88		
9	2067618	Homely 3- Feel at hor	10575274 t	f		1	Whitehorse	-37.8276	145.1175	Entire townhouse	Entire rental unit	6	2 baths		3	["Self check in", "Sauna", "Heating", "Elevator"]	\$207.50	2	365	77 t	13	#####	13/6/2026	5	5	5	2.02		
10	2079904	Modern 1- â ... Stunn	10394006 t	t		14	Melbourne	-37.8166	144.9533	Entire rental unit	Entire rental unit	4	1 bath		2	["Sauna", "Heating", "Elevator"]	\$192.12	2	1125	46 t	374	6/1/2014	#####	4.83	4.89	4.7	2.47		
11	2084207	Apartment With its m	976168 t	f		1	Yarra	-37.8046	144.9832	Entire rental unit	Entire rental unit	2	1 bath		1	["Heating", "Elevator"]	\$85.71	7	21	22 t	100	26/1/2014	7/1/2026	4.96	4.97	4.93	0.66		
12	2099179	Reduced-r Cat lover r	2831931 t	f		1	Port Phillip	-37.859	144.9984	Entire rental unit	Entire rental unit	2	1 bath		1	["Self check in", "Long term stays allc"]	\$85.71	7	21	22 t	100	26/1/2014	7/1/2026	4.96	4.97	4.93	0.66		
13	2106830	Wahroong I have ope	10741024 t	f		1	Melbourne	-37.7962	144.9684	Entire rental unit	Entire rental unit	6	1.5 baths		3	["Long term stays allc"]	\$175.00	2	1125	284 t	563	#####	7/6/2026	4.96	4.92	4.9	4.33		
14	2119988	Sleep me Great Spa	10810140 t	t		3	Melbourne	-37.8093	144.9598	Entire rental unit	Entire rental unit	2	1 bath		1	["Self check in", "Long term stays allc"]	\$175.00	2	1125	284 t	563	#####	7/6/2026	4.96	4.92	4.9	4.33		
15	2124756	Large One Melbourne	10836093 t	f		1	Port Phillip	-37.855	144.9815	Entire rental unit	Entire rental unit	2	1 bath		1	["Long term stays allc"]	\$87.00	1	180	259 t	73	13/4/2014	28/5/2026	4.71	4.85	4.74	0.49		
16	2128253	Sensation! Perfectly s	8908174 t	f		1	Port Phillip	-37.8626	144.9746	Entire rental unit	Entire rental unit	2	1 bath		1	["Long term stays allc"]	\$240.00	3	730	220 t	85	24/1/2014	9/4/2026	4.93	4.8	4.86	0.56		
17	2136008	Stylish Wa Welcome!	1444621 t	f		1	Yarra	-37.8035	144.9858	Entire rental unit	Entire rental unit	2	1.5 baths		1	["Self check in", "First aid kit", "Outlet covers", "Iron"]	\$346.97	2	100	52 t	165	18/1/2014	#####	4.94	4.95	4.91	1.09		
18	2142770	â† Award This fully s	10936267 t	f		1	Port Phillip	-37.863	144.9877	Entire guesthouse	Entire rental unit	4	1 bath		2	["Outlet covers", "Iron"]	\$346.97	2	100	52 t	165	18/1/2014	#####	4.94	4.95	4.91	1.09		
19	2150936	Queen bed Walking di	10857350 t	f		5	Wyndham	-37.8926	144.6678	Private room	Private room	2	2 baths		3	["Self check in", "Long term stays allc"]	\$87.00	1	180	259 t	73	13/4/2014	28/5/2026	4.71	4.85	4.74	0.49		
20	2156314	Spacious - Stay in you	3007990 t	f		2	Port Phillip	-37.8582	144.9693	Entire rental unit	Entire rental unit	4	1 bath		1	["Self check in", "First aid kit", "Outlet covers", "Iron"]	\$240.00	3	730	220 t	85	24/1/2014	9/4/2026	4.93	4.8	4.86	0.56		
21	2157135	Quiet, clea Convenien	11008407 t	f		1	Stonnington	-37.8524	145.0103	Private room	Private room	1	1 shared bath		1	["First aid kit", "Outlet covers", "Iron"]	\$80.80	5	31	11 t	32	20/10/2013	30/4/2026	4.88	4.94	4.77	0.23		
22	2178434	Large Mell Looking fo	11111107 t	t		1	Maribyrnong	-37.807	144.9019	Entire rental unit	Entire rental unit	15	3 baths		9	12	["Self check in", "Garden view"]	\$953.00	3	1125	242 t	160	24/2/2014	19/4/2026	4.84	4.75	4.77	1.07	
23	2194291	Lovely Brk In the mid	11191682 t	f		1	Yarra Range	-37.7505	145.3117	Private room	Private room	1	1 private bath		1	["Garden view"]	\$81.00	2	6	89 t	15	30/12/201	7/4/2024	4.2	4.13	4.53	0.11		
24	2234455	BEACHSID Modern aj	3013970 t	f		1	Bayside	-37.933	144.9999	Entire rental unit	Entire rental unit	2	1.5 baths		1	["Long term stays allc"]	\$240.00	7	1125	0 t	42	29/4/2014	16/12/201	4.83	4.71	4.69	0.28		
25	2234483	Abbotsford's All-roun	11377202 t	f		1	Yarra	-37.803	144.9951	Entire rental unit	Entire rental unit	4	1 bath		2	["Heating", "Washer"]	\$240.00	2	1125	0 t	11	28/1/2014	24/11/201	5	5	5	0.07		
26	2238738	Nook Mell Welcome!	9510598 t	t		56	Melbourne	-37.8244	144.9603	Entire rental unit	Entire rental unit	6	2 baths		2	2	["Self check in", "Washer"]	\$292.67	2	365	96 t	280	7/2/2014	1/5/2026	4.72	4.71	4.73	1.86	
27	2245554	Beautiful (WELCOMI)	11468856 t	f		1	Melbourne	-37.818	144.9875	Private room	Private room	2	1 private bath		1	["Self check in", "Heating", "Smoke alarm"]	\$86.33	4	1125	43 t	316	6/2/2014	26/4/2026	4.71	4.73	4.76	2.1		
28	2258577	Art Deco g Our beaut	2657687 t	f		2	Port Phillip	-37.8629	144.9808	Entire rental unit	Entire rental unit	4	1 bath		2	["Heating", "Smoke alarm"]	\$86.33	2	5	0 t	15	13/3/2014	17/3/2015	4.57	4.38	4.46	0.1		
29	2263264	Light & Sp Our light h	11557695 t	f		1	Port Phillip	-37.8823	144.9794	Entire rental unit	Entire rental unit	4	1 bath		2	["Heating", "Hair dryer"]	\$253.60	1	1125	199 t	531	15/5/2014	16/6/2026	4.79	4.84	4.74	3.61		
30	2292052	Luxury Ma Converted	10621363 t	t		3	Port Phillip	-37.8599	144.9751	Entire loft	Entire rental unit	2	1 bath		1	["Self check in", "Shared bathroom"]	\$253.60	1	1125	199 t	531	15/5/2014	16/6/2026	4.79	4.84	4.74	3.61		
31	2301999	Hush â- Hi The Hush	11755700 t	t		5	Yarra Range	-37.862	145.3595	Entire cottage	Entire rental unit	2	1 bath		1	["Self check in", "Shared bathroom"]	\$480.00	1	20	342 t	264	#####	8/5/2026	4.55	4.68	4.53	1.85		
32	2357769	An extra! Exterior w	12041542 t	f		1	Port Phillip	-37.8618	144.9815	Entire rental unit	Entire rental unit	4	1.5 baths		2	2	["Shared bathroom"]	\$571.00	2	1125	314 t	63	22/4/2014	23/1/2026	4.73	4.43	4.74	0.43	
33	2361740	Stunning, Gorgeous	638404 t	t		1	Yarra	-37.8032	144.9843	Entire rental unit	Entire rental unit	4	1 bath		2	2	["TV with internet", "Iron", "Shampoo", "Washer"]	\$293.40	1	1125	165 t	60	19/8/2014	8/3/2026	4.97	4.92	4.93	0.42	
34	2362450	Renovated Recently r	10985248 t	f		1	Glen Eira	-37.8901	145.0662	Entire rental unit	Entire rental unit	6	2 baths		3	["Iron", "Shampoo", "Washer"]	\$293.40	4	1125	0 t	4	13/5/2014	30/3/2015	4.75	5	4.5	0.03		

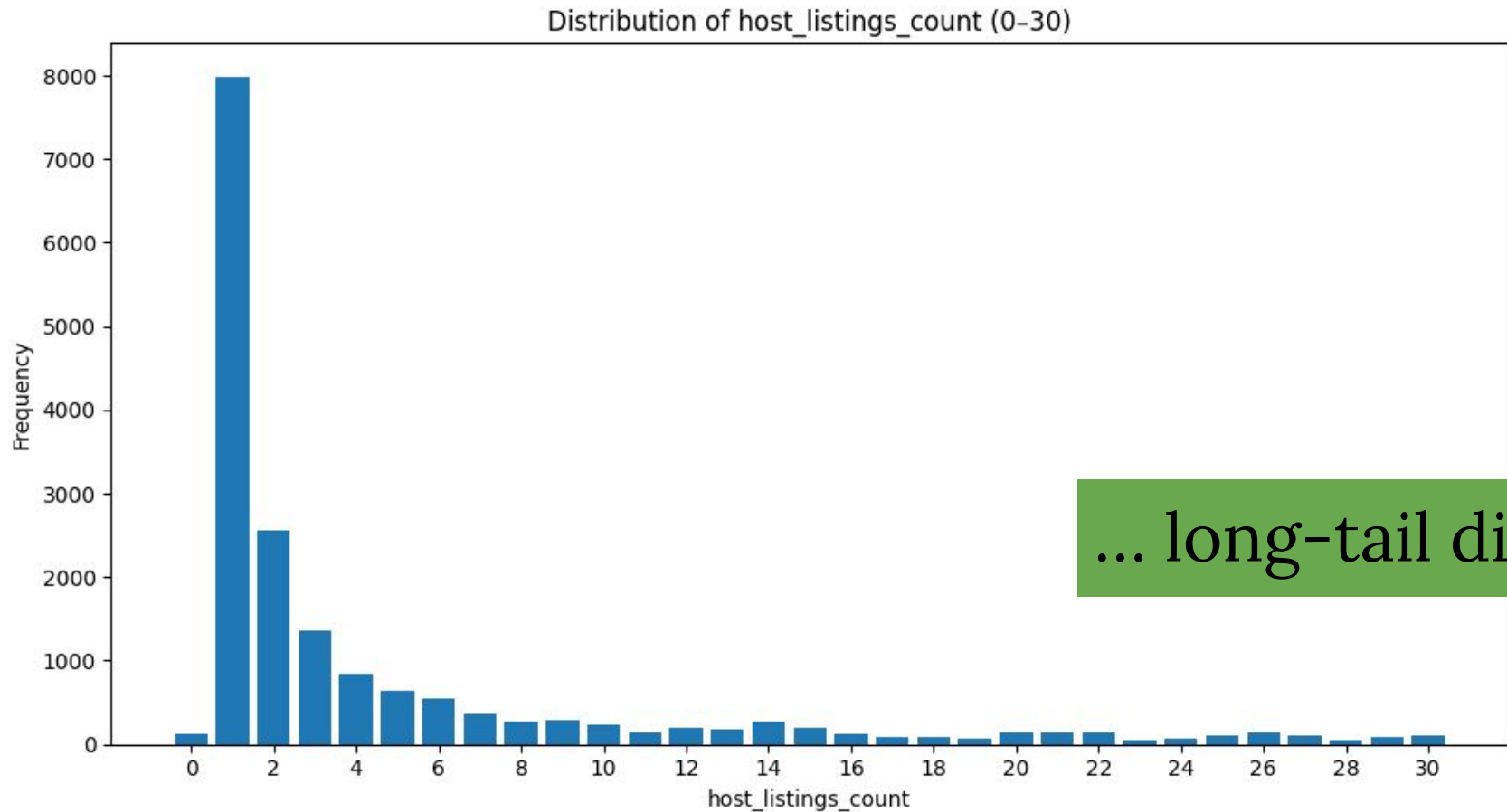
Recap: distributions

A1: AirBnB property listings

host_listings_count

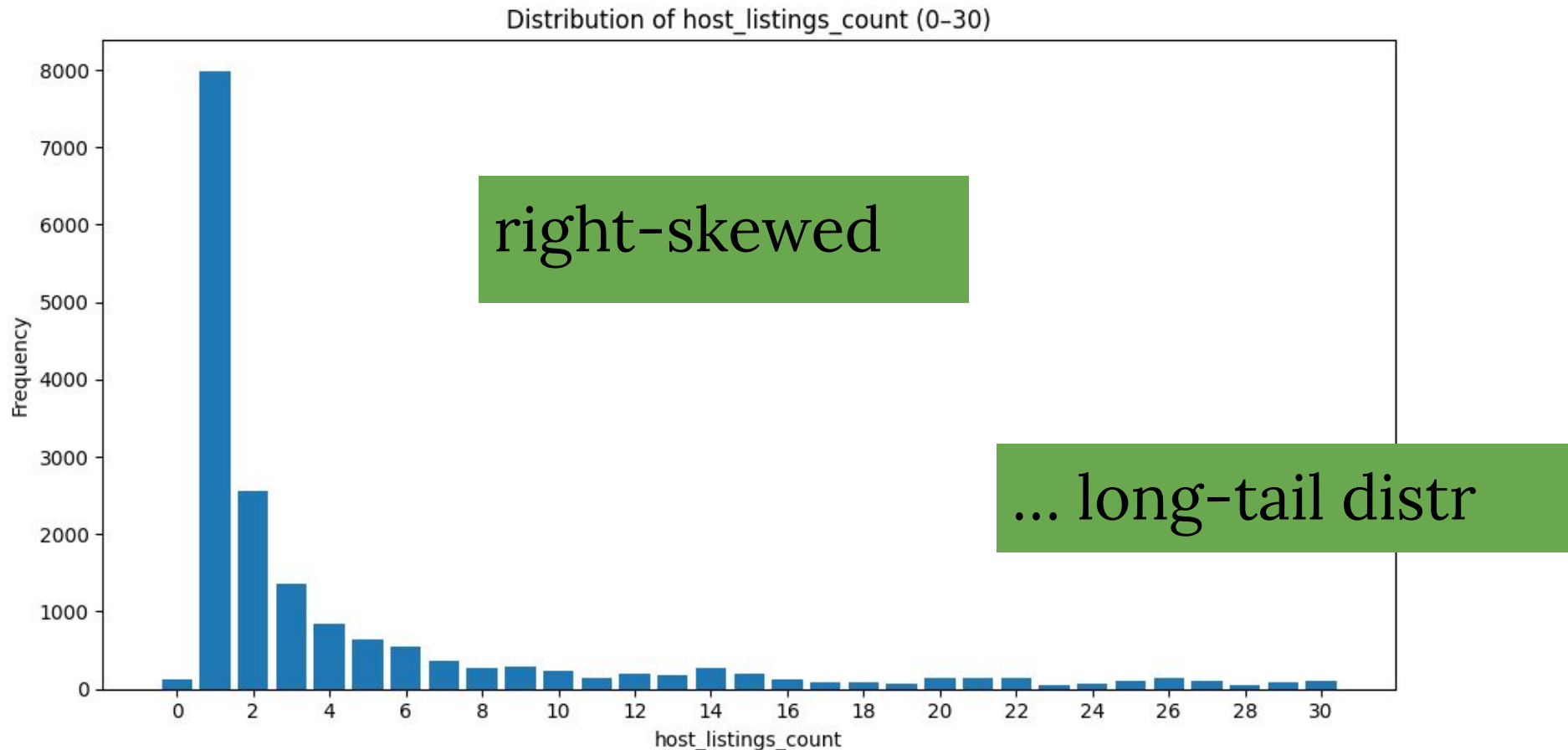
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC
1	id	name	description	host_id	host_iden	host_is	host_listin	neighbour	latitude	longitude	property_	room_tpy	accommod	bathroom	bedrooms	beds	amenities	price	minimum	maximum	availability	has_availa	number_o	first_revie	last_review	review_score	review_score	review_score	reviews_per
2	10803	Room in D A large air	38901 t	f	1	Moreland	-37.7661	144.9795	Private roc	Private roc	2	1 shared bath	1	1	1	1	["Self che	\$60.74	5	28	158 t	214	#####	4/5/2026	4.57	4.07	4.66	1.31	
3	12936	St Kilda 18 RIGHT IN T	50121 t	f	10	ort Philli	-37.86	144.9766	Entire rent	Entire hon	2	1 bath	1	1	1	1	["Dishes and silverwa	3	14	0 t	42	4/8/2010	15/3/2020	4.68	4.71	4.66	0.22		
4	41836	CLOSE TO Easy to tra	182833 t	f	3	arebin	-37.6976	145.0007	Private roc	Private roc	2	1 shared bath	1	1	1	1	["Heating", "Hair drye	7	1125	0 t	157	16/11/2012	22/8/2018	4.71	4.65	4.69	0.83		
5	43429	Tranquil Ja No service	189684 t	t	4	Monash	-37.8998	145.1158	Entire rent	Entire hon	2	1 bath	1	1	1	1	["Self che	\$165.00	2	1125	278 t	280	#####	14/5/2026	4.88	4.94	4.86	1.48	
6	44699	15 yearsH Unwaverir	189245 t	f	3	ort Philli	-37.8316	144.9706	Private roc	Private roc	2	1 shared bath	1	1	1	1	["Bay view	\$215.00	1	14	169 t	67	9/9/2010	17/5/2026	4.77	4.83	4.71	0.35	
7	47100	Cosy, cute Our home	212071 t	f	2	arra	-37.8184	145.005	Entire rent	Entire hon	2	1 bath	1	1	1	1	["Long term stays allc	4	21	0 t	180	9/9/2010	8/1/2025	4.88	4.92	4.81	0.94		
8	51592	Central Cit Stay in a st	190879 t	t	2	Melbourn	-37.8127	144.9631	Entire loft	Entire hon	2	1.5 baths	1	1	1	1	["Self che	\$294.10	2	1125	208 t	537	22/2/2011	7/6/2026	4.91	4.75	4.79	2.88	
9	2067618	Homely 3- Feel at hor	10575274 t	f	1	Whitehors	-37.8276	145.1175	Entire tow	Entire hon	6	2 baths	3	3	3	3	["Self che	\$207.50	2	365	77 t	13	#####	13/6/2026	5	5	5	2.02	
10	2079904	Modern 1- â ... Stunn	10394006 t	t	14	Melbourn	-37.8166	144.9533	Entire rent	Entire hon	4	1 bath	1	2	1	2	["Sauna", "I	\$192.12	2	1125	46 t	374	6/1/2014	#####	4.83	4.89	4.7	2.47	
11	2084207	Apartment With its m	976168 t	f	1	arra	-37.8046	144.9832	Entire rent	Entire hon	2	1 bath	1	1	1	1	["Heating", "Elevator	3	30	0 t	10	14/2/2016	16/7/2017	4.9	4.6	4.56	0.08		
12	2099179	Reduced-r Cat lover r	2831931 t	f	1	ort Philli	-37.859	144.9984	Entire rent	Entire hon	2	1 bath	1	1	1	1	["Self che	\$85.71	7	21	22 t	100	26/1/2014	7/1/2026	4.96	4.97	4.93	0.66	
13	2106830	Wahroong I have ope	10741024 t	f	1	Melbourn	-37.7962	144.9684	Entire hon	Entire hon	6	1.5 baths	3	1	1	1	["Long term stays allc	6	1125	0 t	178	13/8/2014	#####	4.93	4.87	4.81	1.23		
14	2119988	Sleep me Great Spac	10810140 t	t	3	Melbourn	-37.8093	144.9598	Entire rent	Entire hon	2	1 bath	1	1	1	1	["Self che	\$175.00	2	1125	284 t	563	#####	7/6/2026	4.96	4.92	4.9	4.33	
15	2124756	Large One Melbourn	10836093 t	f	1	ort Philli	-37.855	144.9815	Entire rent	Entire hon	2	1 bath	1	1	1	1	["Long term stays allc	1	1125	0 t	34	18/3/2014	22/7/2023	4.79	4.76	4.73	0.23		
16	2128253	Sensation: Perfectly s	8908174 t	f	1	ort Philli	-37.8626	144.9746	Entire rent	Entire hon	2	1 bath	1	1	1	1	["Long term stays allc	3	200	0 t	30	3/1/2016	25/11/201	4.97	5	4.93	0.23		
17	2136008	Stylish Wa Welcome	1444621 t	f	1	arra	-37.8035	144.9858	Entire rent	Entire hon	2	1.5 baths	1	1	1	1	["Self che	\$346.97	2	100	52 t	165	18/1/2014	#####	4.94	4.95	4.91	1.09	
18	2142770	â† Award This fully s	10936267 t	f	1	ort Philli	-37.863	144.9877	Entire gue	Entire hon	4	1 bath	1	2	1	2	["Outlet covers", "Iro	2	1125	0 t	1	20/1/2014	20/1/2014	5	5	5	0.01		
19	2150936	Queen bey Walking di	10857350 t	f	5	Vyndham	-37.8926	144.6678	Private roc	Private roc	2	2 baths	1	3	1	3	["Self che	\$87.00	1	180	259 t	73	13/4/2014	28/5/2026	4.71	4.85	4.74	0.49	
20	2156314	Spacious - Stay in you	3007990 t	f	2	ort Philli	-37.8582	144.9693	Entire rent	Entire hon	4	1 bath	1	1	1	1	["Self che	\$240.00	3	730	220 t	85	24/1/2014	9/4/2026	4.93	4.8	4.86	0.56	
21	2157135	Quiet, clea Convenien	11008407 t	f	1	tonningtc	-37.8524	145.0103	Private roc	Private roc	1	1 shared bath	1	1	1	1	["First aid	\$80.80	5	31	11 t	32	20/10/2013	30/4/2026	4.88	4.94	4.77	0.23	
22	2178434	Large Mell Looking fo	11111107 t	t	1	Maribyrnc	-37.807	144.9019	Entire hon	Entire hon	15	3 baths	9	12	1	12	["Self che	\$953.00	3	1125	242 t	160	24/2/2014	19/4/2026	4.84	4.75	4.77	1.07	
23	2194291	Lovely Brk In the mid	11191682 t	f	1	arra Rang	-37.7505	145.3117	Private roc	Private roc	1	1 private b	1	1	1	1	["Garden v	\$81.00	2	6	89 t	15	30/12/201	7/4/2024	4.2	4.13	4.53	0.11	
24	2234455	BEACHSID Modern aj	3013970 t	f	1	ayside	-37.933	144.9999	Entire rent	Entire hon	2	1.5 baths	1	1	1	1	["Long term stays allc	7	1125	0 t	42	29/4/2014	16/12/201	4.83	4.71	4.69	0.28		
25	2234483	Abbotsford's All-roun	11377202 t	f	1	arra	-37.803	144.9951	Entire hon	Entire hon	4	1 bath	2	1	1	1	["Heating", "Washer"	2	1125	0 t	11	28/1/2014	24/11/201	5	5	5	0.07		
26	2238738	Nook Mell Welcome	9510598 t	t	56	Melbourn	-37.8244	144.9603	Entire rent	Entire hon	6	2 baths	2	2	1	2	["Self che	\$292.67	2	365	96 t	280	7/2/2014	1/5/2026	4.72	4.71	4.73	1.86	
27	2245554	Beautiful (WELCOMI	11468856 t	f	1	Melbourn	-37.818	144.9875	Private roc	Private roc	2	1 private bath	1	1	1	1	["Self che	\$86.33	4	1125	43 t	316	6/2/2014	26/4/2026	4.71	4.73	4.76	2.1	
28	2258577	Art Deco g Our beaut	2657687 t	f	2	ort Philli	-37.8629	144.9808	Entire rent	Entire hon	4	1 bath	2	1	1	1	["Heating", "Smoke a	2	5	0 t	15	13/3/2014	17/3/2015	4.57	4.38	4.46	0.1		
29	2263264	Light & Sp Our light f	11557695 t	f	1	ort Philli	-37.8823	144.9794	Entire rent	Entire hon	4	1 bath	2	1	1	1	["Heating", "Hair drye	2	14	0 t	7	30/12/2014	24/4/2016	4.57	4.57	4.57	0.05		
30	2292052	Luxury Ma Converted	10621363 t	t	3	ort Philli	-37.8599	144.9751	Entire loft	Entire hon	2	1 bath	1	1	1	1	["Self che	\$253.60	1	1125	199 t	531	15/5/2014	16/6/2026	4.79	4.84	4.74	3.61	
31	2301999	Hush â- Hi The Hush	11755700 t	t	5	arra Rang	-37.862	145.3595	Entire cott	Entire hon	2	1 bath	1	1	1	1	["Self che	\$480.00	1	20	342 t	264	#####	8/5/2026	4.55	4.68	4.53	1.85	
32	2357769	An extrads Exterior w	12041542 t	f	1	ort Philli	-37.8618	144.9815	Entire hon	Entire hon	4	1.5 baths	2	2	1	2	["Shared b	\$571.00	2	1125	314 t	63	22/4/2014	23/1/2026	4.73	4.43	4.74	0.43	
33	2361740	Stunning, Gorgeous	838404 t	t	1	arra	-37.8032	144.9843	Entire rent	Entire hon	4	1 bath	2	2	1	2	["TV with	\$293.40	1	1125	165 t	60	19/8/2014	8/3/2026	4.97	4.92	4.93	0.42	
34	2362450	Renovated Recently r	10985248 t	f	1	len Eira	-37.8901	145.0662	Entire hon	Entire hon	6	2 baths	3	1	1	1	["Iron", "Shampoo",	4	1125	0 t	4	13/5/2014	30/3/2015	4.75	5	4.5	0.03		

Recap: histogram of listings_count

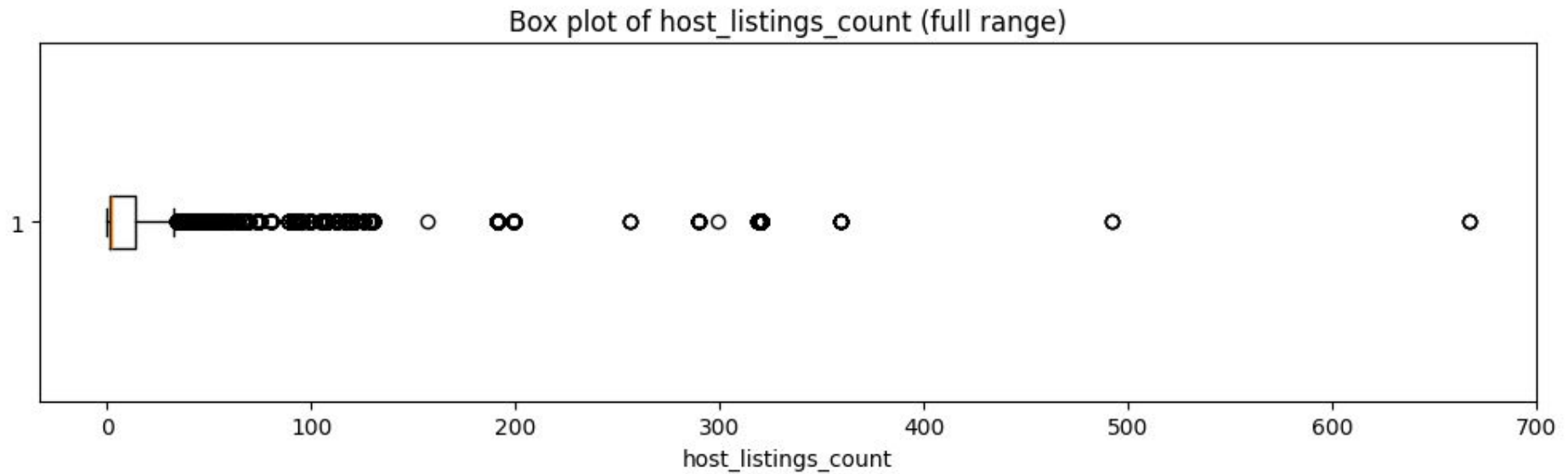


... long-tail distr

Recap: histogram of listings_count

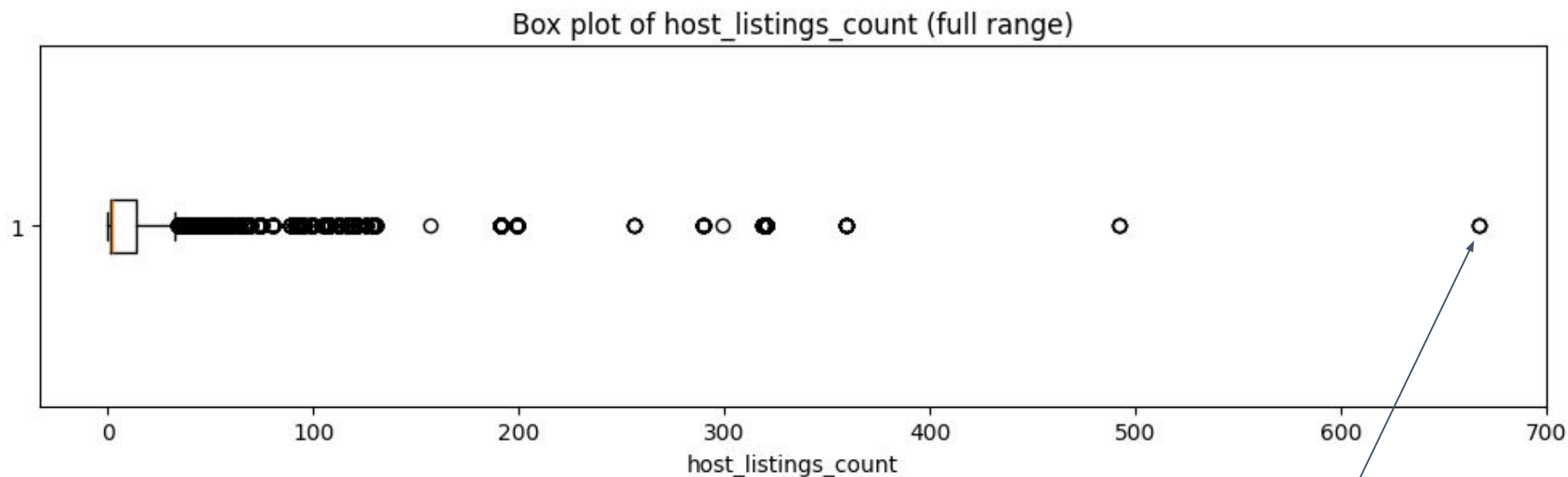


Recap: box plot of listings_count



Minimum	0
Q1 (25th percentile)	1
Median (Q2)	2
Mean	20.68
Q3 (75th percentile)	14
Maximum	667
IQR = Q3 - Q1	13
Mode	1

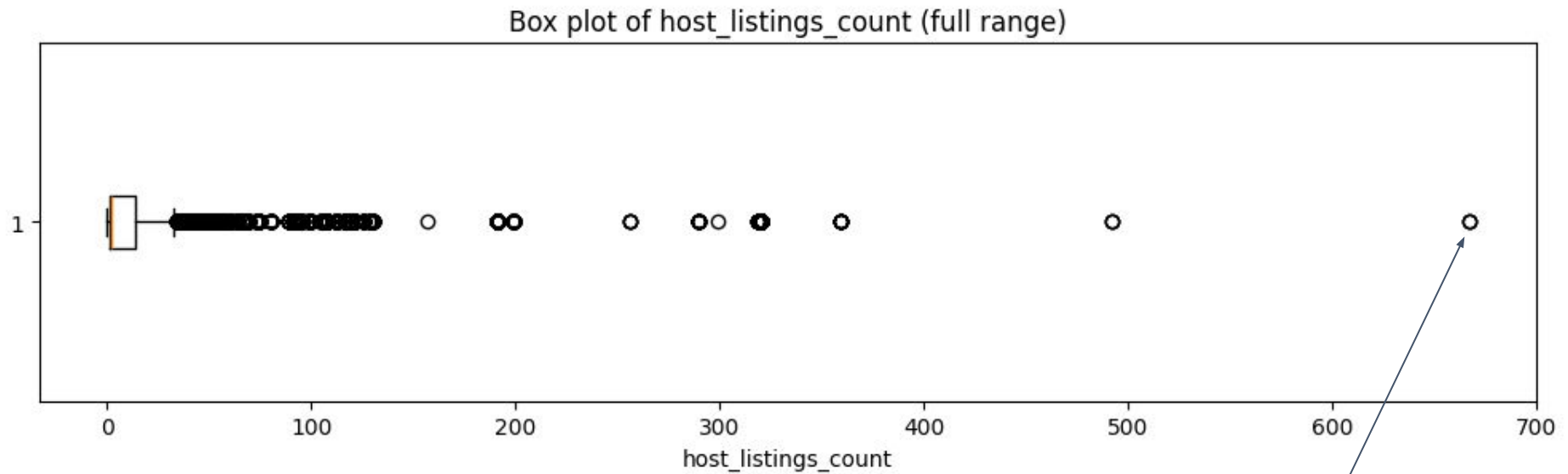
Recap: box plot of listings_count



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!

Recap: box diagram of listings_count



Why?

Minimum	0
Q1 (25th percentile)	1
Median (Q2)	2
Mean	20.68
Q3 (75th percentile)	14
Maximum	667
IQR = Q3 - Q1	13
Mode	1

!

Data in higher dimensions 3x95

	ambience	apartment	art	bathroom	bedroom	building	ceiling	city	d
doc1	0.1771	0.0000	0.1771	0.0000	0.0000	0.0000	0.1347	0.0000	
doc2	0.0000	0.2895	0.0000	0.1101	0.0000	0.2895	0.0000	0.1101	
doc3	0.0000	0.0000	0.0000	0.2229	0.2931	0.0000	0.1114	0.2229	

Difficult to **visualize data** when **each item has lots of features**

How can we automatically determine what the **significant groups / segments / communities of the data are?**

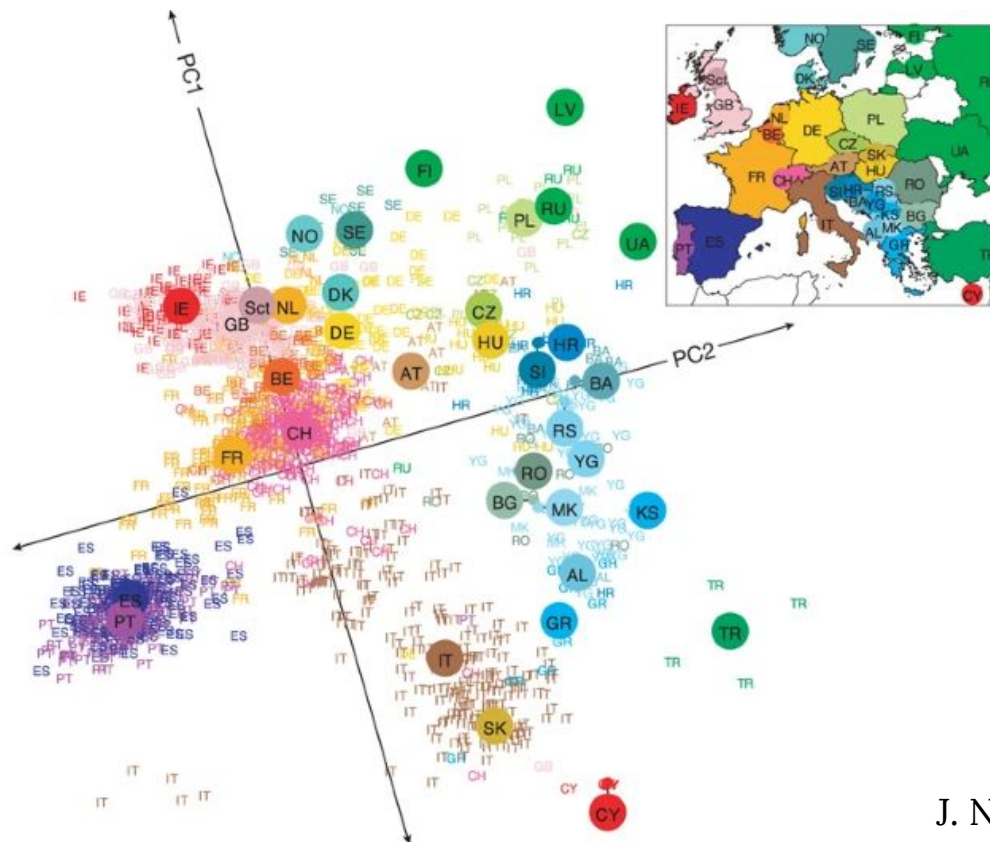
- If we can, then:
 - We can understand the data better
 - We can apply separate interventions to each group (e.g. marketing campaign)

A couple of Inspiring Examples

Genes mirror geography within Europe

Which countries resemble each other?
Why?

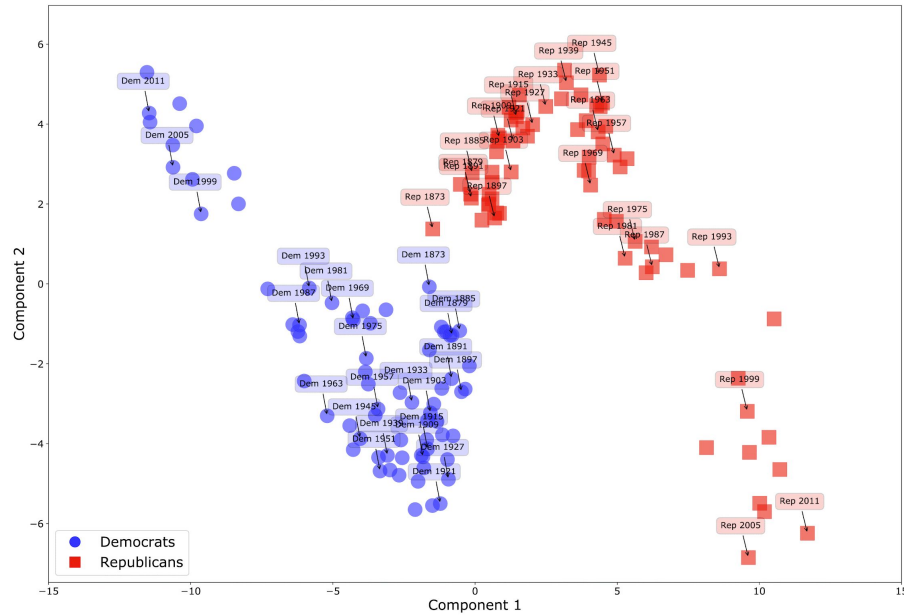
researchers studied the genomes of 3,192 European individuals at 500,568 loci (positions across the genome)



J. Novembre et al., *Nature* 000, 1-4 (2008)

Parliament Speeches

(a) Two-Dimensional Projection

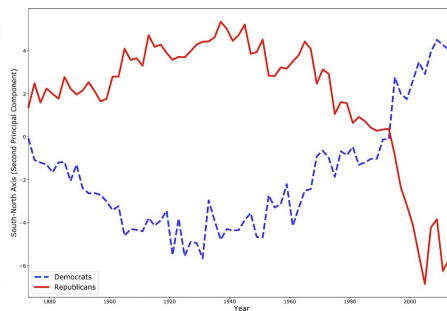


Party placement in the US House (1873–2016)

(b) First Dimension



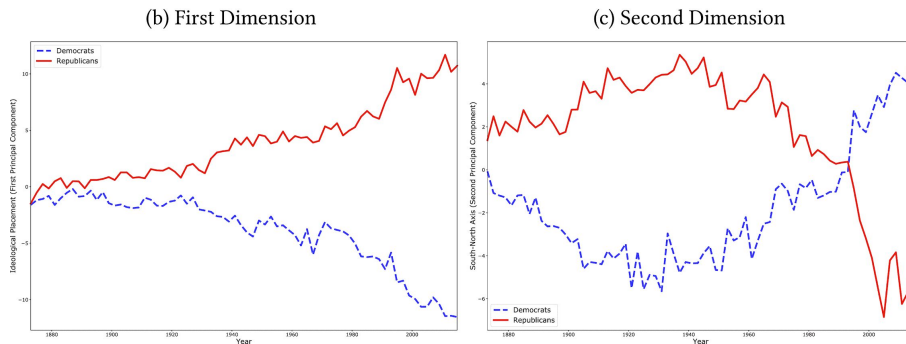
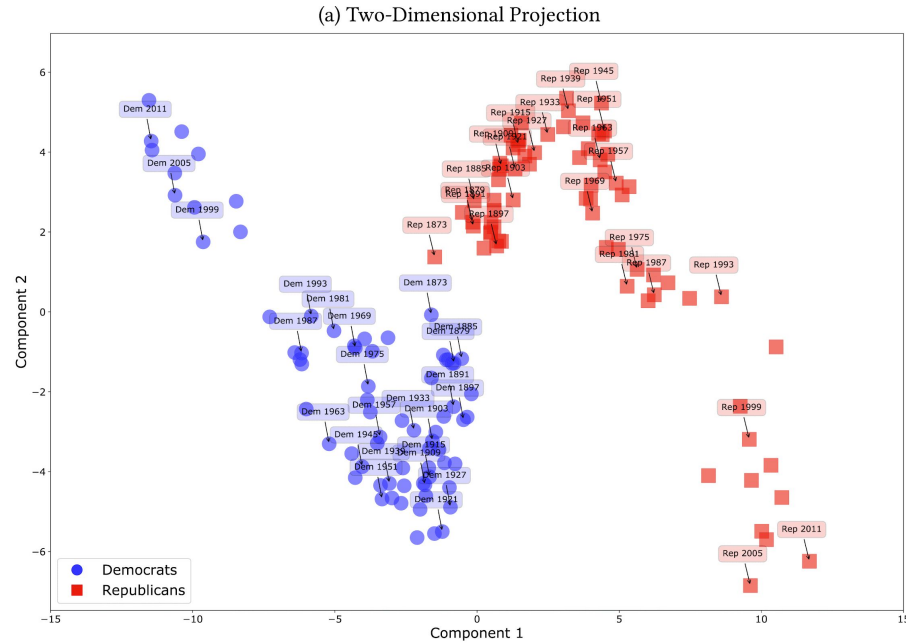
(c) Second Dimension



<https://www.cambridge.org/core/journals/political-analysis/article/word-embeddings-for-the-analysis-of-ideological-placement-in-parliamentary-corpora/017F0CEA9B3DB6E1B94AC36A509A8A7B>

Parliament Speeches

Party placement in the US House (1873–2016)



- Grouping samples into clusters
- Projecting samples into 2D/3D

<https://www.cambridge.org/core/journals/political-analysis/article/word-embeddings-for-the-analysis-of-ideological-placement-in-parliamentary-corpora/017F0CEA9B3DB6E1B94AC36A509A8A7B>

Lecture outline

1. **General introduction**
2. **k-Means clustering**
3. Visualisation of clustering tendency (VAT)
4. Hierarchical clustering (Part 2)
5. Dimensionality reduction (Part 2)

Applications of clustering

Clustering is a major task in **data analysis and visualisation**

- Market segmentation
- Image analysis
- Document Clustering
- Outlier detection

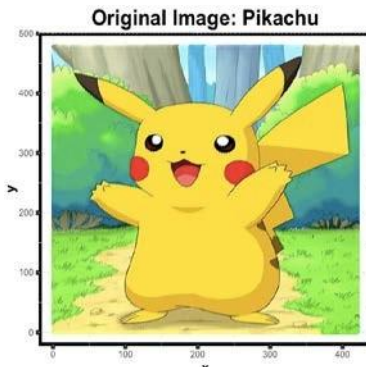
Clustering helps us do two things:

- Find meaningful groups in the data
- Avoid being overwhelmed by too much detail

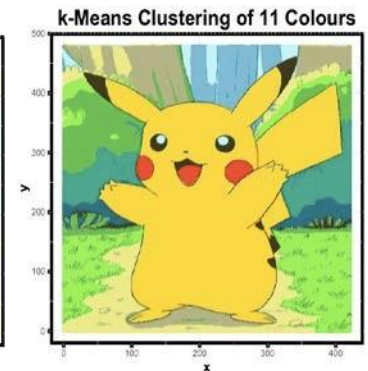
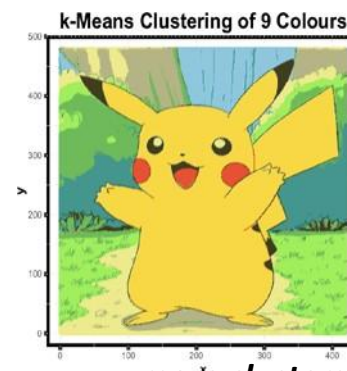
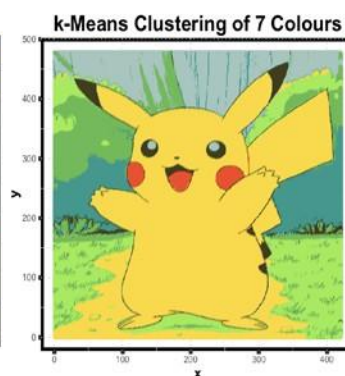
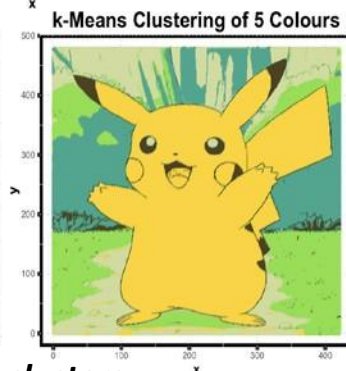
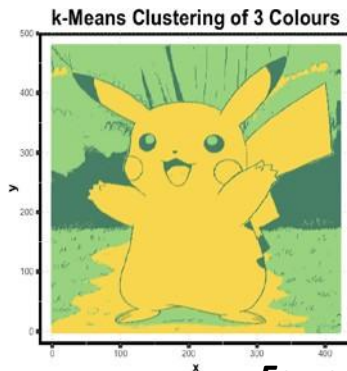
Applications of clustering

Clustering Applications: Image Discretization

- Reassign colour (RGB) value to k distinct colours
- **Clustering (Grouping) pixels** using colour difference, *not spatial data*



 $[248, 221, 45] \rightarrow [245, 215, 50]$
 $[250, 218, 40] \rightarrow [245, 215, 50]$
 $[241, 214, 55] \rightarrow [245, 215, 50]$



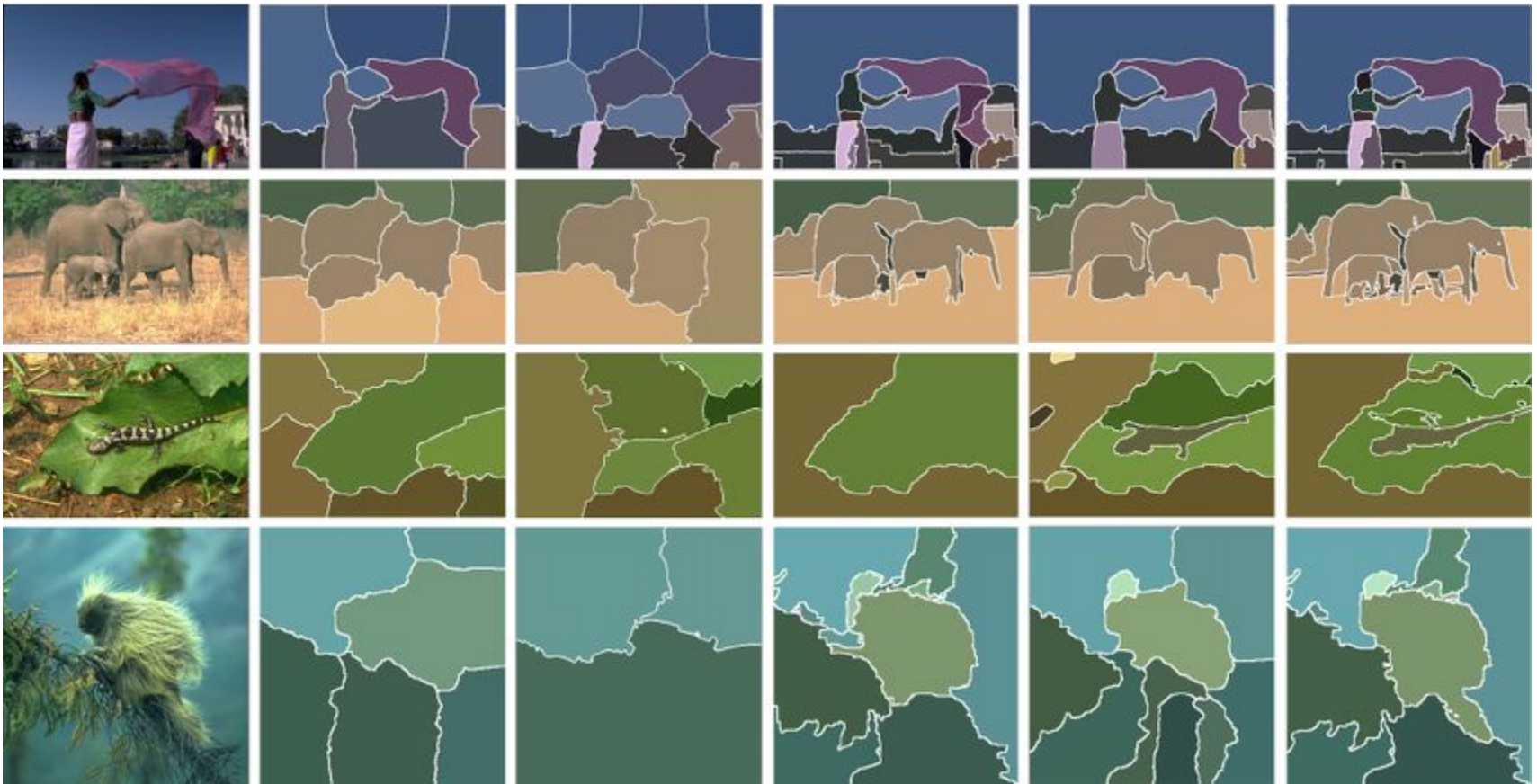
Fewer clusters

more clusters

Applications of clustering

Clustering Applications: Image Segmentation

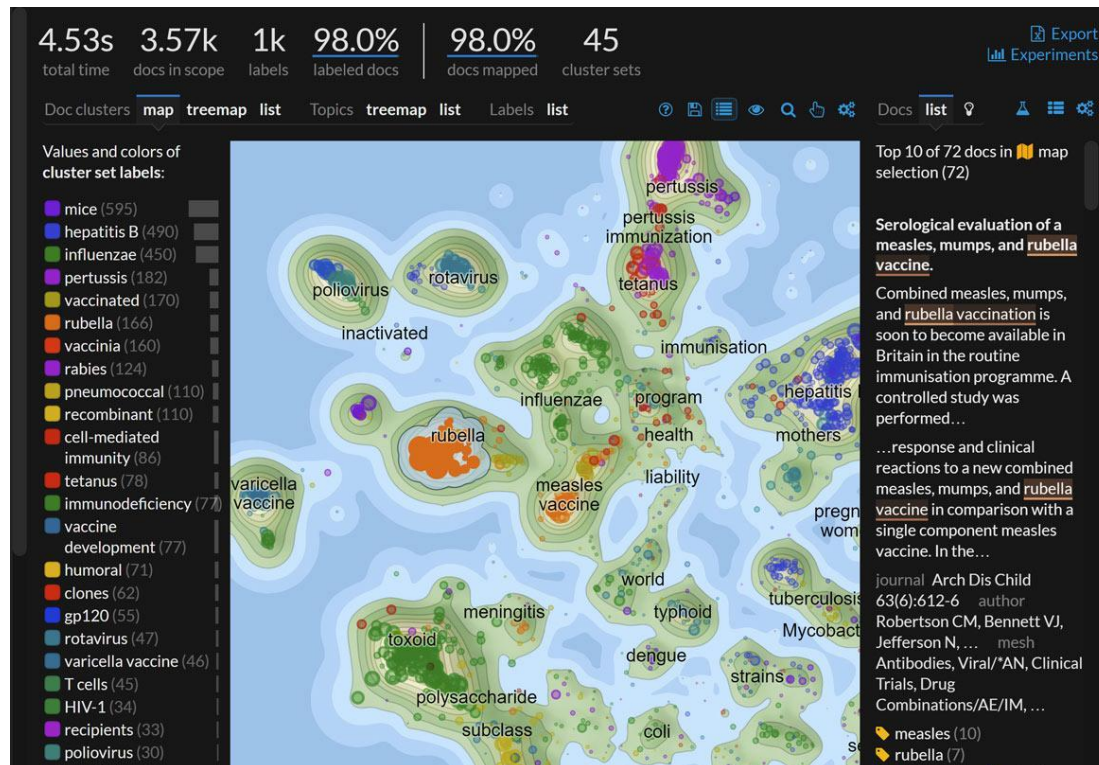
- *Partition a digital image into multiple segments*



Applications of clustering

Clustering Applications: Document Clustering

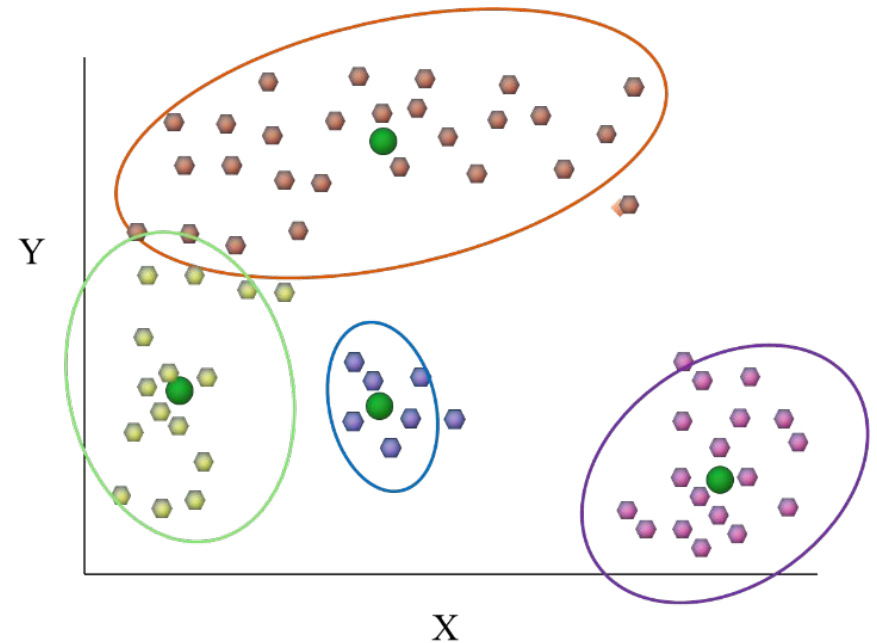
- 2D map of PubMed medical-paper abstracts: Assign textual documents to different topic clusters by using terms or other features



<https://lingo-dev.datascience.columbia.edu/apps/explorer/>

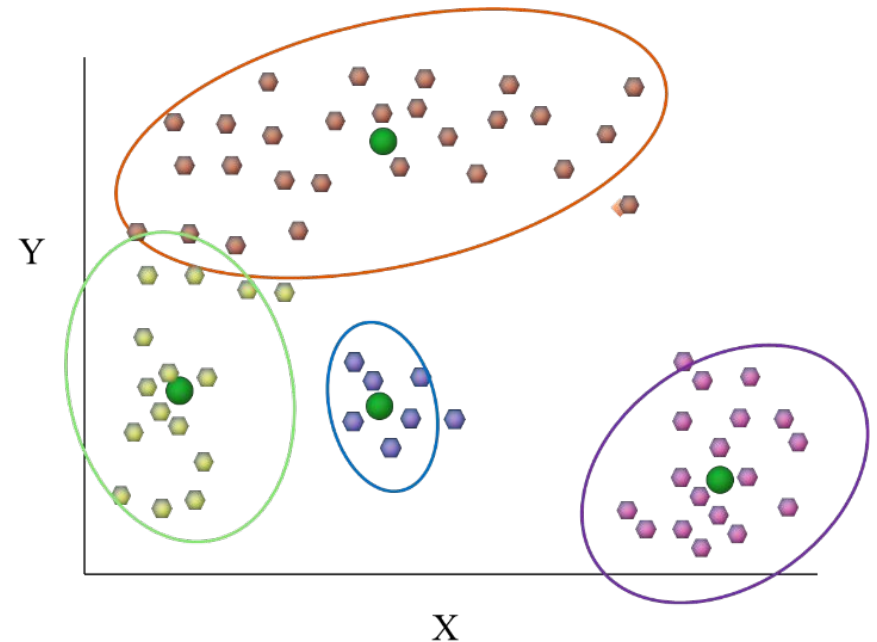
Three desiderata for clusters

1. We must assign each object to exactly one cluster
2. We want similar objects in a cluster and different objects in different clusters
3. We want each cluster to be summarised by its **centroid** (the 'average' of all its objects)



Three desiderata for clusters

1. We must assign each object to exactly one cluster
2. We want similar objects in a cluster and different objects in different clusters
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Requires: – similarity measure
– adjusting the number of clusters
→ assessment the cluster quality

Cluster analysis

- There are many clustering algorithms
- They all try to achieve the desiderata
- They all eventually use some sort of similarity/distance metric

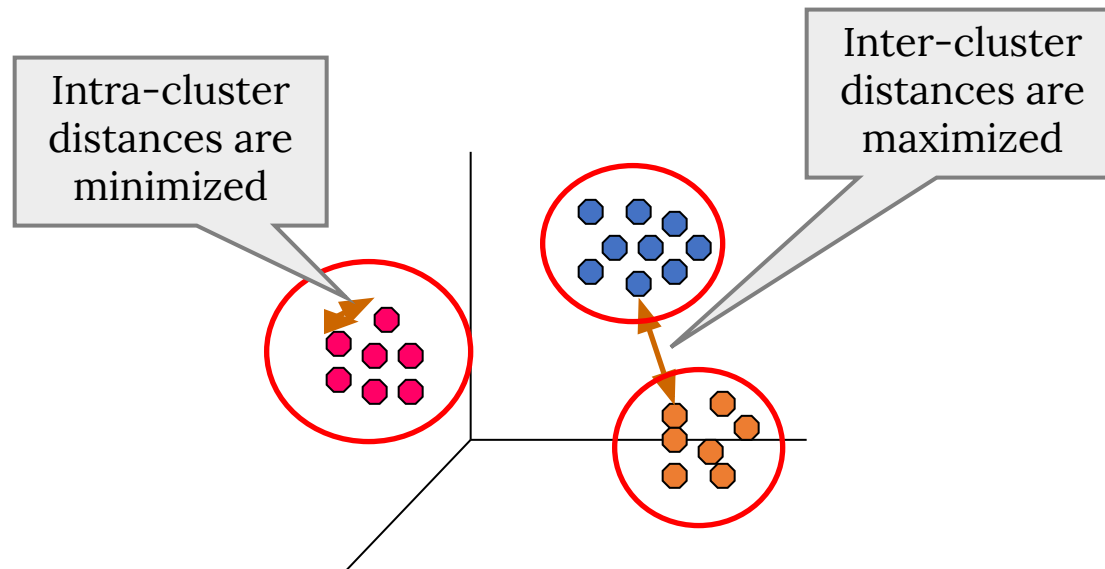


Figure from Tan, Steinbach and Kumar 2004

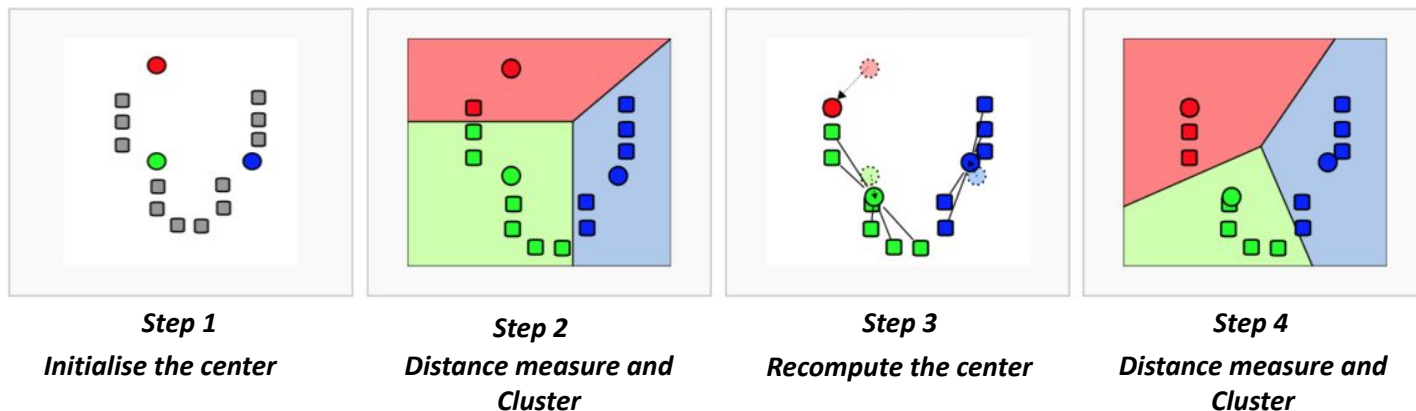
Partitioning Clustering

Partitioning clustering are clustering methods that require the analyst to specify **the number of clusters to be generated**.

how many clusters we want?

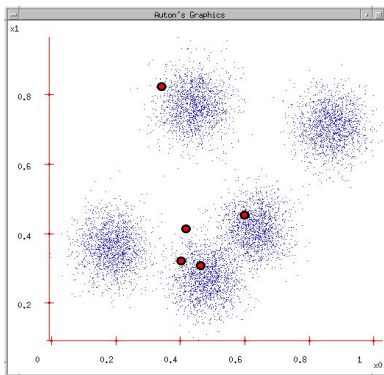
The best-known family of partitioning clustering algorithms is **k-means clustering**.

Demonstration of the standard K-means clustering algorithm

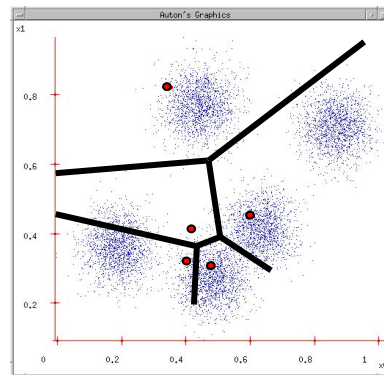


k-Means Clustering

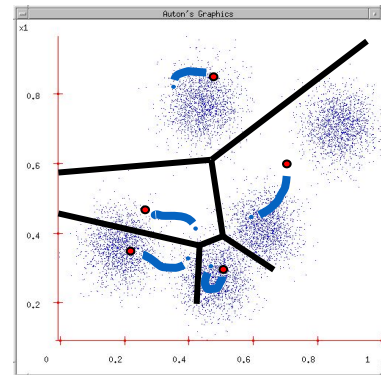
1. Ask user how many clusters they would like (e.g. $k=5$)
2. Randomly pick k cluster centroids
3. For each data point find out which centroid it is closest to. Thus each centroid “owns” a set of datapoints
4. Each cluster finds the new centroid of the points in it
5. New centroids => new boundaries
6. Repeat steps 3–5 until no further changes



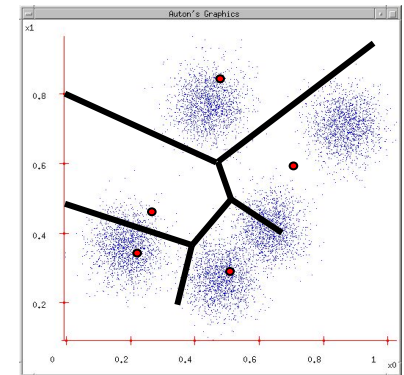
Procedure 1,2



Procedure 3



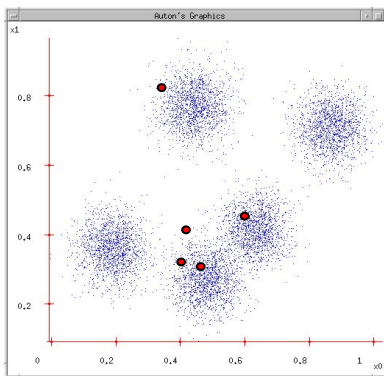
Procedure 4



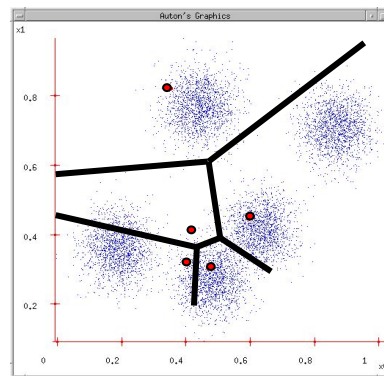
Procedure 5

k-Means Clustering

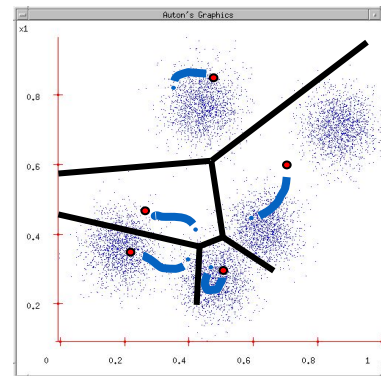
1. Ask user how many clusters they would like (e.g. $k=5$)
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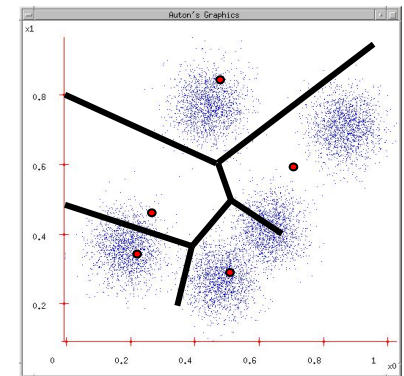
Procedure 1,2



Procedure 3



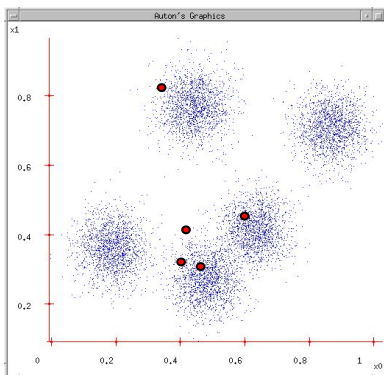
Procedure 4



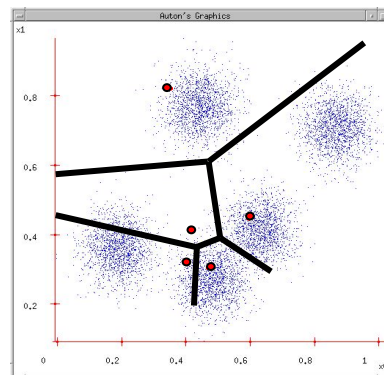
Procedure 5

k-Means Clustering

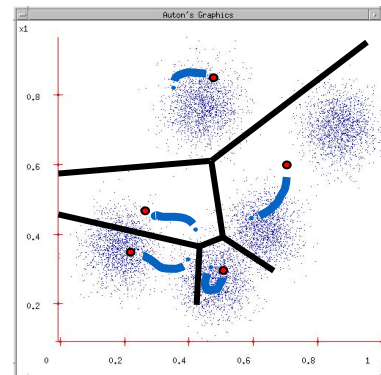
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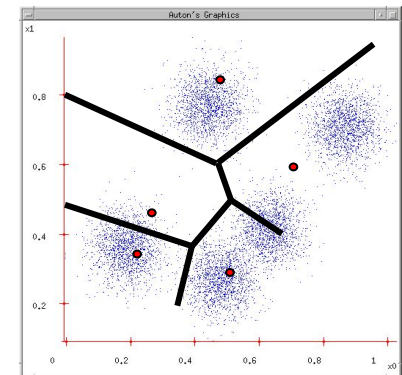
Procedure 1,2



Procedure 3



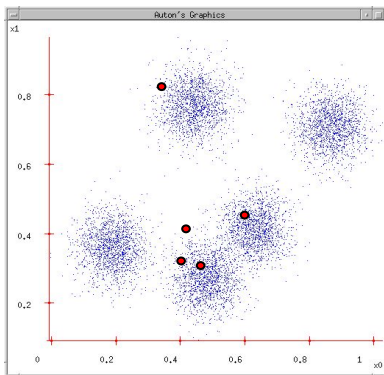
Procedure 4



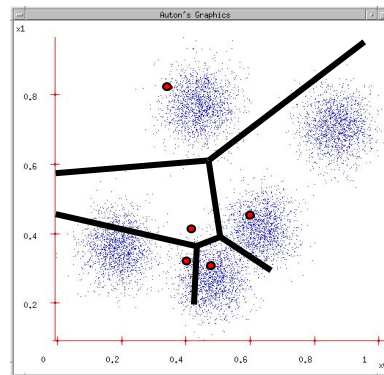
Procedure 5

k-Means Clustering

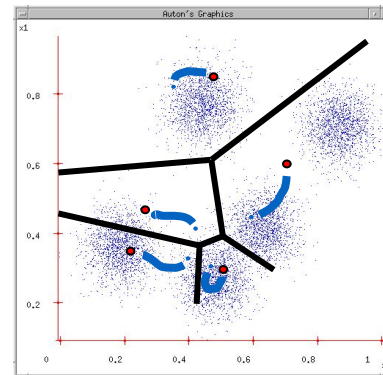
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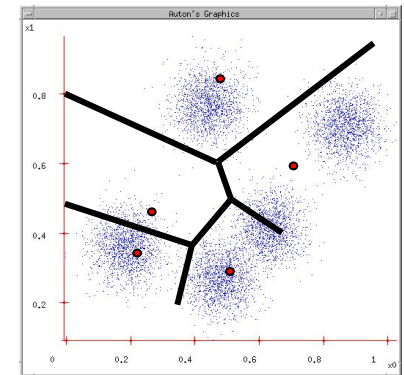
Procedure 1,2



Procedure 3



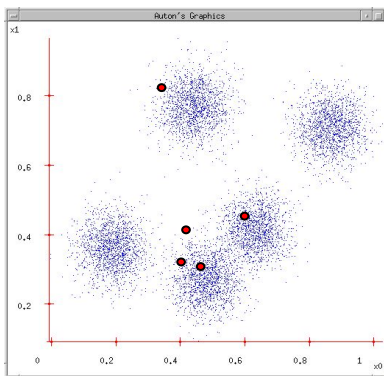
Procedure 4



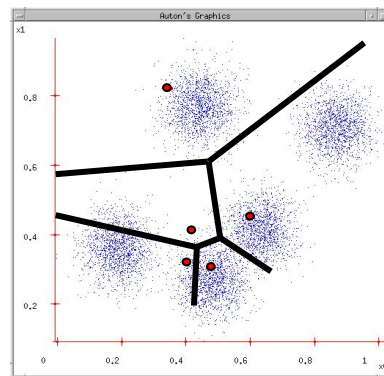
Procedure 5

k-Means Clustering

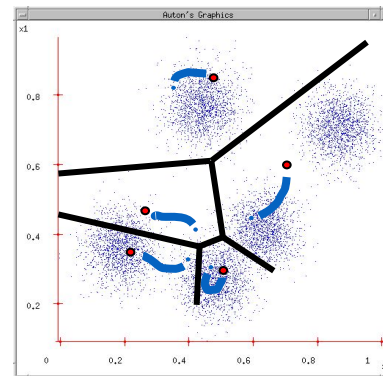
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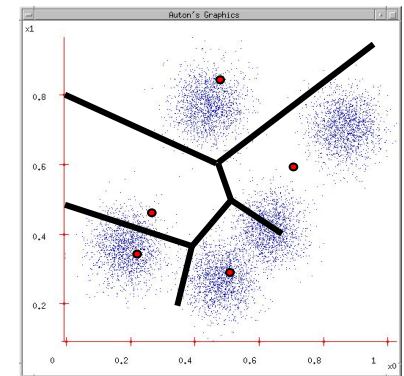
Procedure 1,2



Procedure 3



Procedure 4



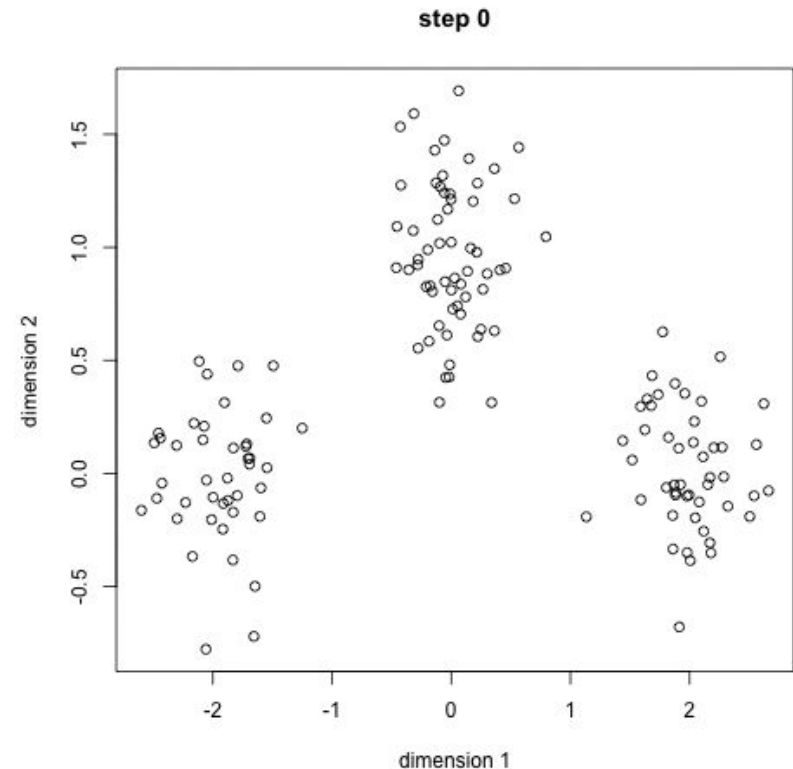
Procedure 5

k-Means Algorithm Summary

Input: set of datapoints and parameter k

Algorithm:

1. Randomly select k datapoints as the seeds. These are the starting clusters
 2. Assign each point to the cluster whose centroid is closest to it (the centroid is the **mean point** = centre of the cluster)
 3. Compute the centroids of the clusters that have now formed
 4. Go back to step 2; stop repeating when nothing changes
- Note that a datapoint can 'hop' from cluster to cluster depending on what other points join the cluster



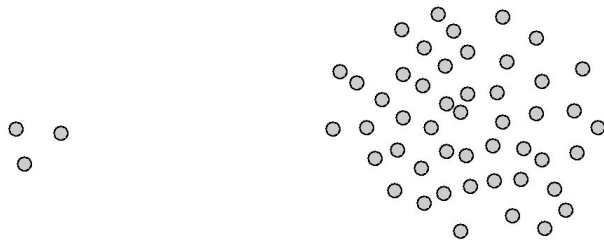
EM = Expectation Maximization

Understanding the algorithm

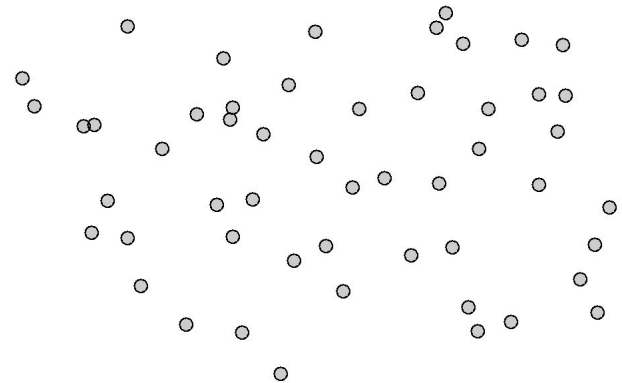
For which dataset does k-Means require the fewest iterations?

$k=2$

Dataset 1



Dataset 2



k-Means details

- Typically choose the initial seed points randomly
 - Different runs of the algorithm will produce different results
- Closeness usually measured by **Euclidean distance**
 - Can also use other distance functions (e.g. correlation)

Given $X = (x_1, x_2, x_3, \dots, x_n)$ and $Y = (y_1, y_2, y_3, \dots, y_n)$

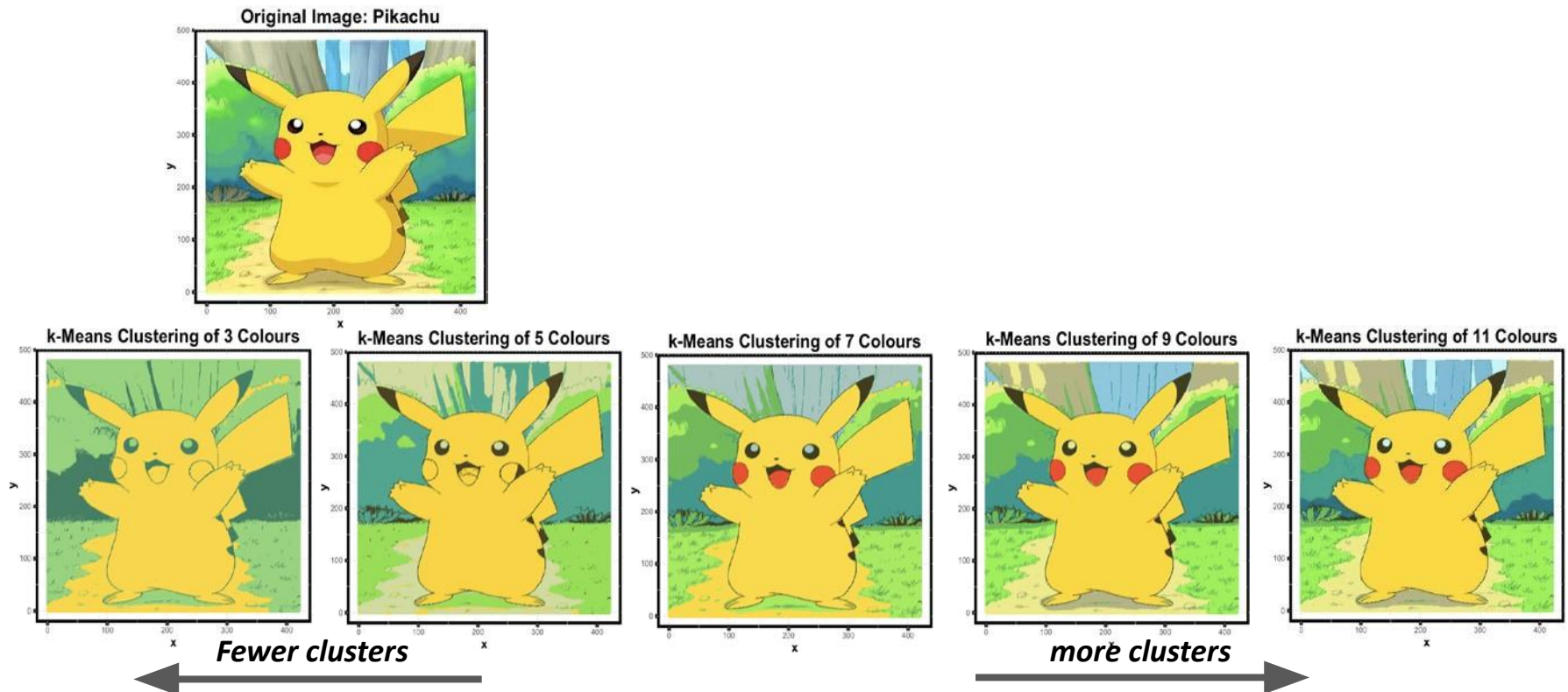
$$D(X, Y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + (x_3 - y_3)^2 + \dots (x_n - y_n)^2}$$

- Algorithm converges to a local optimum — typically does not require many iterations

Applications of clustering (recap)

Clustering Applications: Image Discretization

- Reassign colour value to k distinct colours
- **Clustering (Grouping) pixels** using colour difference, not spatial data



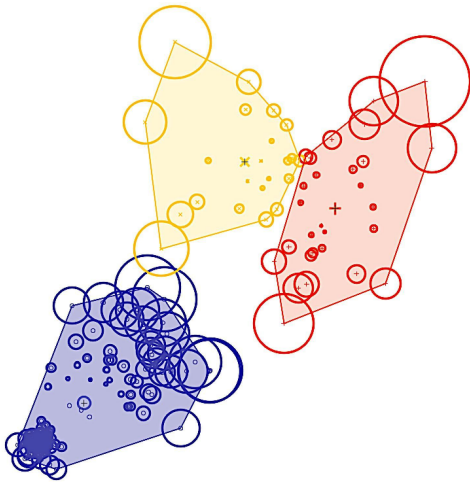
An application: cluster-based outlier detection

Goal: Find the ‘odd cases’ — outliers

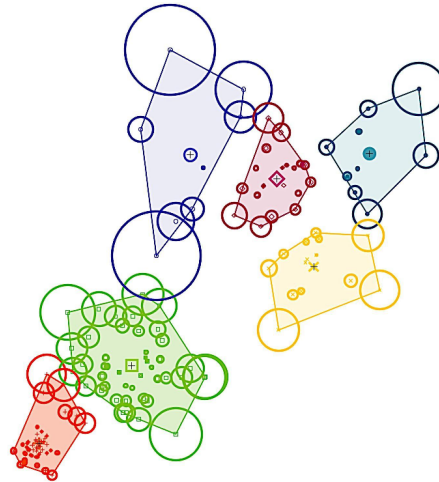
- An *outlier* is far away from any groups of normal objects
- **Each object is associated with exactly one cluster** and its outlier score is equal to **its distance from its cluster centre**
 - outlier score = distance(point, centroid)
- Reward the clustering with the smallest outlier score

Clustering based outlier detection

3 clusters



6 clusters

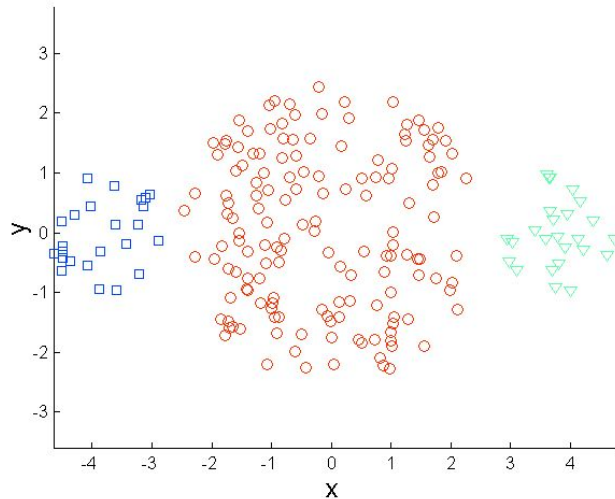


9 clusters

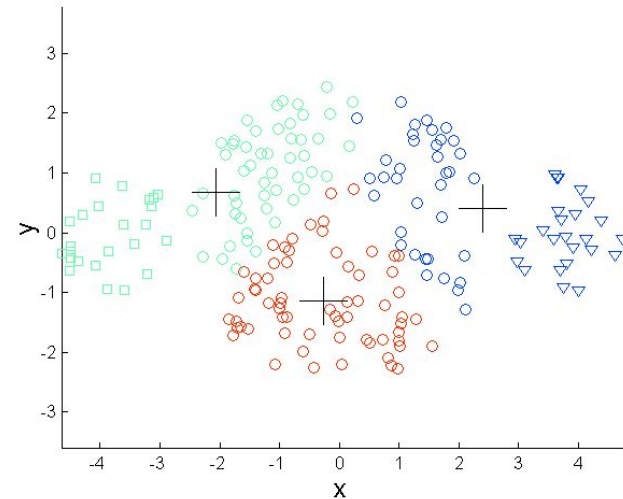


Limitations of K-means(1): Differing Sizes

K-means has problems when clusters are of differing sizes



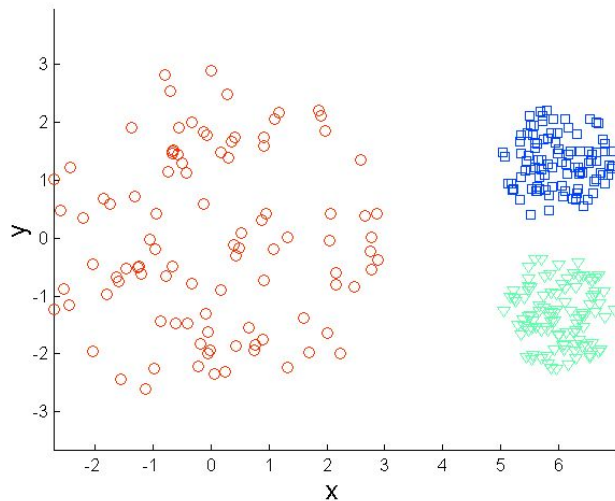
Original Points



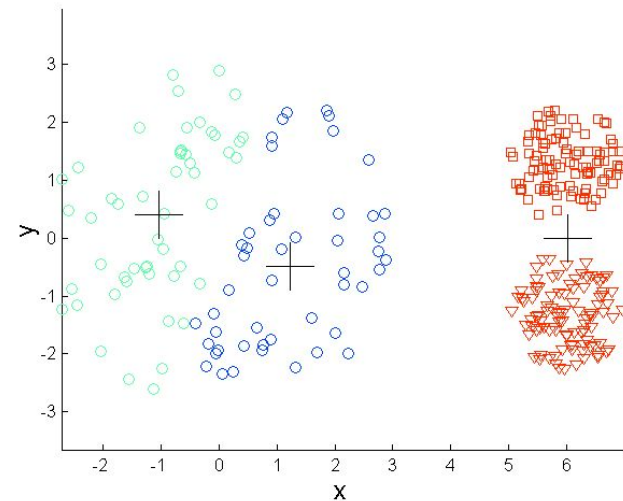
K-means (3 Clusters)

Limitations of K-means(2): Differing Density

K-means has problems when clusters are of differing densities.



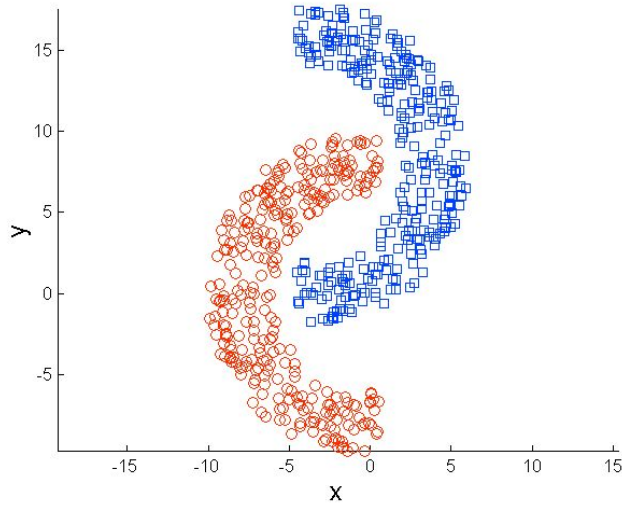
Original Points



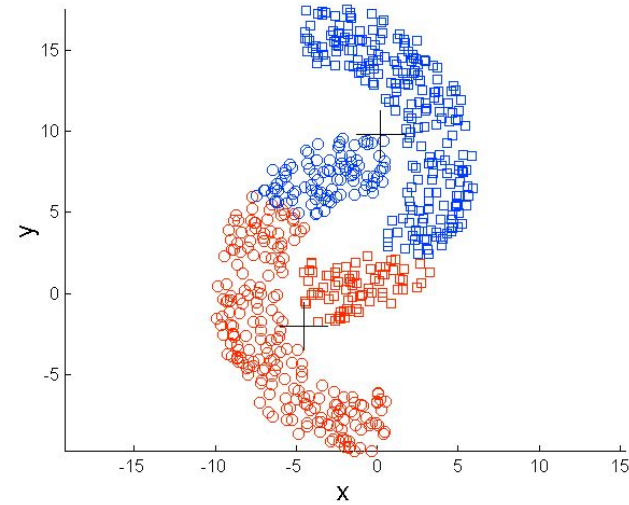
K-means (3 Clusters)

Limitations of K-means(3): Non-globular Shapes

K-means has problems when clusters are with non-globular shapes



Original Points



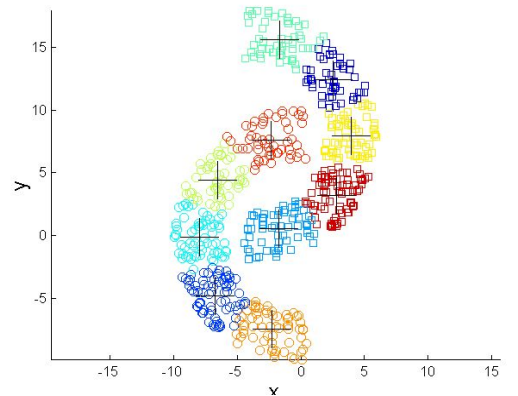
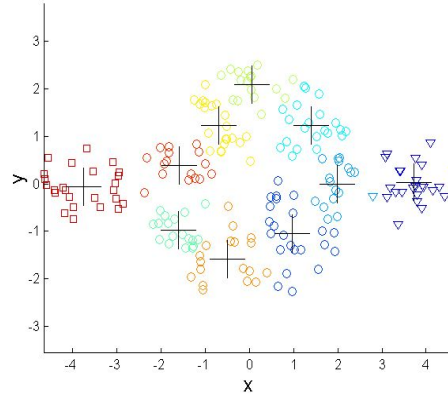
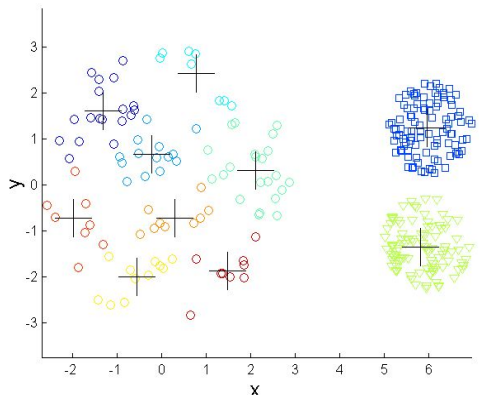
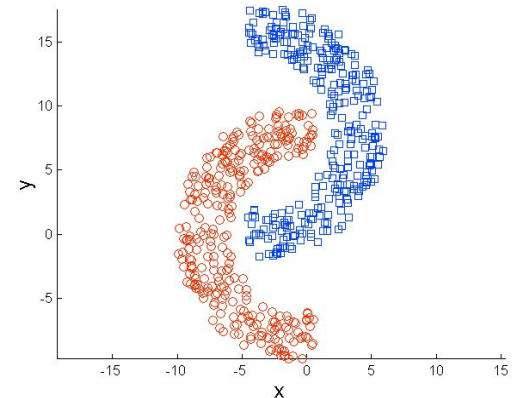
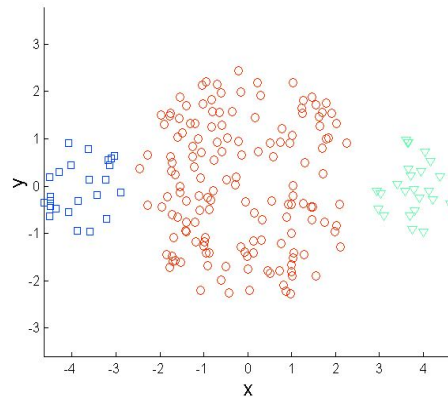
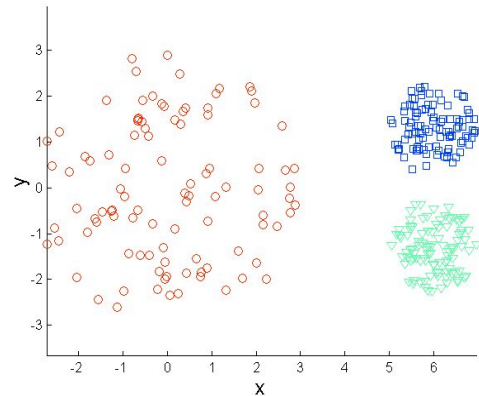
K-means (2 Clusters)

How to overcome those



How to Overcome

An idea: Use many clusters, merge together



Finding k – the elbow method

A good question: What's the best k ?

- Idea: find the k that gives the 'tightest' clusters
- Approach: calculate repeatedly with $k=1, 2, 3...$ and each time measure the '**tightness**'
- Pick the best one

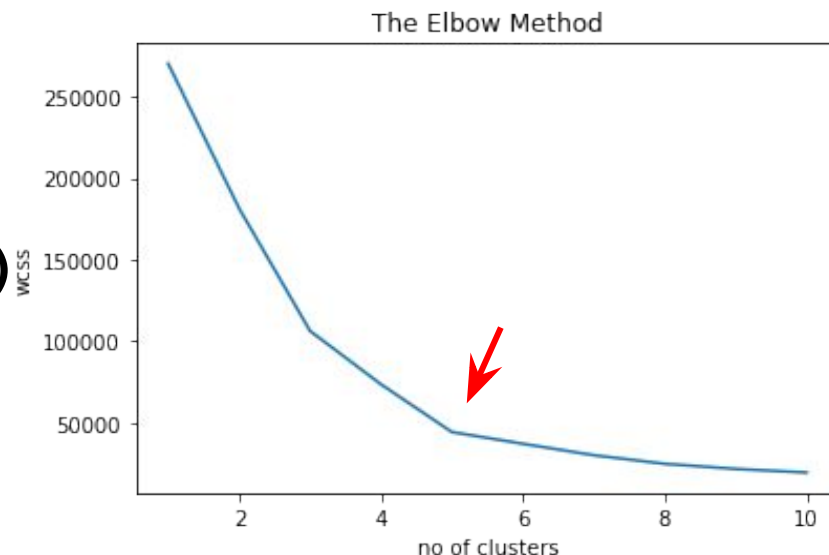
- What measures '**tightness**'?
- Intuition: average distance of each point to the centroid

Finding k – the elbow method

- **‘Tightness’** measure: Within Cluster Sum of Squared distance (WCSS)

$$WCSS(k) = \sum_{c=1}^k \sum_{x_i \in c} \|x_i - \bar{x}_c\|^2, \quad \bar{x}_c \text{ is the centroid of cluster } c$$

- Sum of squared distances of data points to their closest cluster centre
- Called **SSE (Sum of Squared Error)**



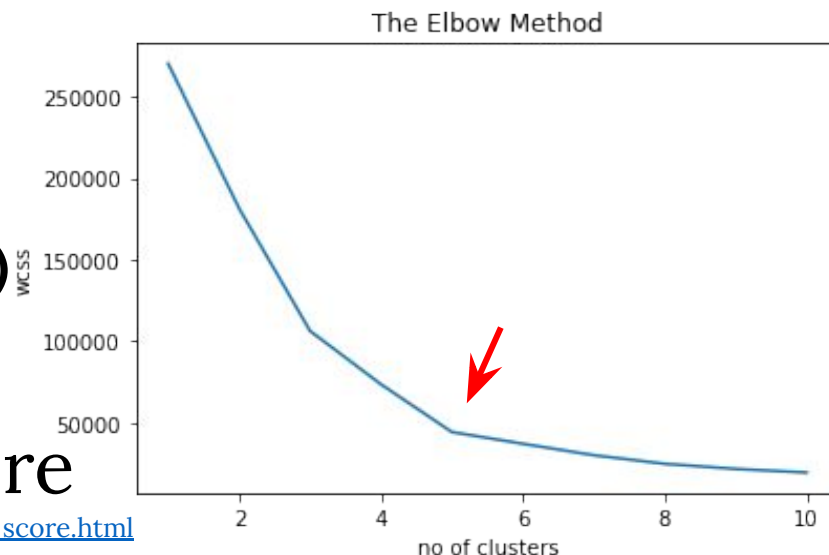
<https://www.kaggle.com/abhishekyadav5/kmeans-clustering-with-elbow-method-and-sil>

Finding k – the elbow method

- **‘Tightness’** measure: Within Cluster Sum of Squared distance (WCSS)

$$WCSS(k) = \sum_{c=1}^k \sum_{x_i \in c} \|x_i - \bar{x}_c\|^2, \quad \bar{x}_c \text{ is the centroid of cluster } c$$

- Sum of squared distances of data points to their closest cluster centre
- Called **SSE (Sum of Squared Error)**



Also common: Silhouette score

https://scikit-learn.org/stable/modules/generated/sklearn.metrics.silhouette_score.html

<https://www.kaggle.com/abhishekyadav5/kmeans-clustering-with-elbow-method-and-sil>

AirBnb Example

200-Sample:

- "property type": Entire townhouse, Entire vacation home, Entire villa, Entire cottage, Entire guest suite, Entire guesthouse, Entire chalet, Entire home
- only where the price is listed
- do stratified sampling across cheap, medium-cost and expensive properties

“Property description”: case-folding, remove punctuation and stop-words, lemmatise

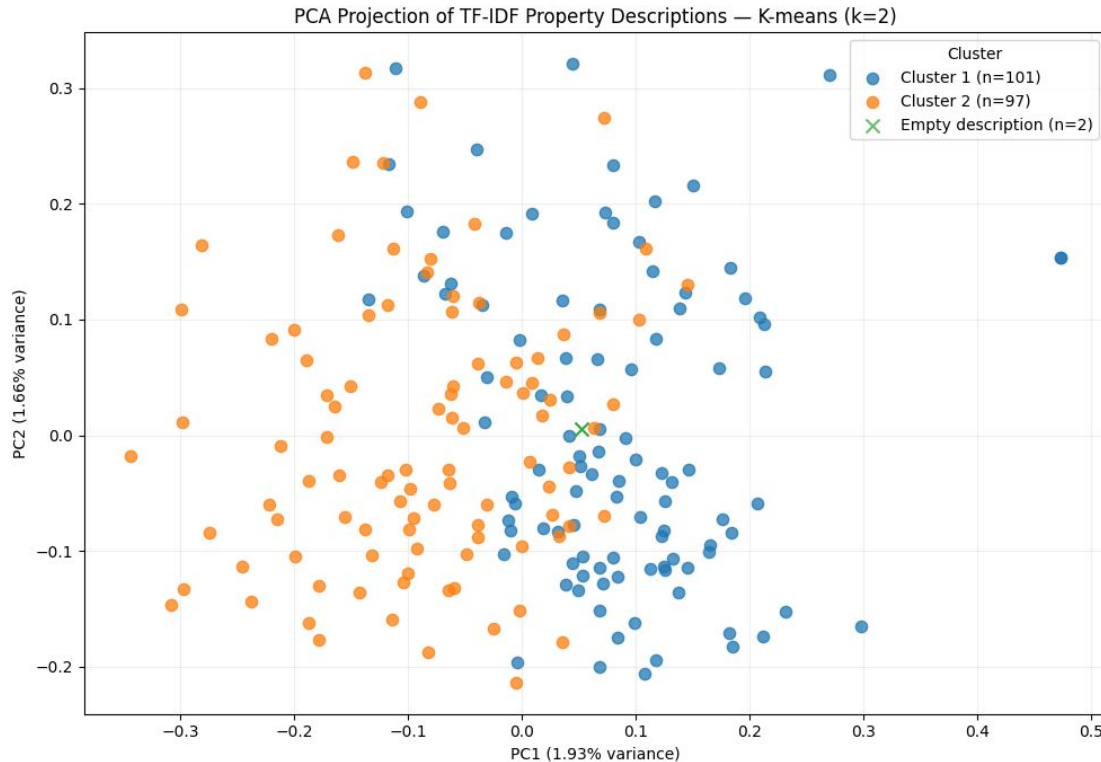
Create term-document matrix using TF-IDF model → **1,641** terms × 200 properties

Dimensionality reduction and k-means with optimising Silhouette score

AirBnb Example

id	name	description	host_id	host_identity_ve	host_is_superho	host_listings_co	neighbourhood	latitude	longitude	property_type	room_type	accommodates	bathrooms_text	bedrooms
1012822797661	Stylish Mordi get	Enjoy a stylish e	324210026	t	f	2	Kingston	-38.00849651	145.0935119	Entire home	Entire home/apt	6	1.5 baths	2
9998677	Stonehill Retreat	Stonehill Retreat	34735288	t	t	2	Nillumbik	-37.6053	145.2444	Entire home	Entire home/apt	4	1 bath	2
45714836	Escape to the cc	Managed by Vall	370200478	t	t	28	Yarra Ranges	-37.8702	145.36257	Entire cottage	Entire home/apt	2	1 bath	1
1679261945530	Ryans Villa	Have fun with th	391647246	t	f	1	Casey	-38.08	145.25244	Entire home	Entire home/apt	10	3 baths	5
5451838425209	Oakleigh unit .	p it is brand new th	385574531	t	f	2	Monash	-37.90193	145.08139	Entire guesthou	Entire home/apt	3	1 bath	2
38935171	Serenity 11 - Wh	LIVE WORK PL	151485462	t	f	3	Cardinia	-38.07758	145.49131	Entire home	Entire home/apt	7	2.5 baths	4
1125558703216	Nice Home Free	WEEK OR MON	340154122	t	f	31	Brimbank	-37.76916	144.83757	Entire home	Entire home/apt	10	1.5 baths	3
1591652770079	4 Bed 1930s Tur	A light-filled, con	738065090	t	f	1	Stonnington	-37.84417	145.01029	Entire home	Entire home/apt	7	3.5 baths	4
7751375071583	30% OFF Spaci	The outstanding	185076876	t	f	20	Maroondah	-37.77947	145.29395	Entire home	Entire home/apt	10	4 baths	5
39803640	Beautiful 2BD At	This 2BD house	3929204	t	f	2	Port Phillip	-37.87518	144.98746	Entire home	Entire home/apt	4	1 bath	2
8398410	Sunny Blue cott	You will feel at h	44246690	t	f	1	Port Phillip	-37.8383522	144.9612274	Entire cottage	Entire home/apt	2	1 bath	2
1344633360431	Brightside Stay	- This bright & spa	16364674	t	t	27	Maribymong	-37.77366042	144.8816231	Entire home	Entire home/apt	6	2 baths	3
1570214077842	Fresh & Modern	Relax with the w	105969199	t	f	1	Melton	-37.73874408	144.6684679	Entire home	Entire home/apt	8	3 baths	4
37884609	Fig Leaf Cottage	The standard pri	126202365	t	t	1	Yarra Ranges	-37.71236	145.37254	Entire cottage	Entire home/apt	4	2 baths	2
8873195933033	Flora Unit - Affo	THIS ACCOMM	514669719	t	t	5	Brimbank	-37.8033741	144.8177511	Entire home	Entire home/apt	4	1 bath	1
20613142	Heart of Fitzroy	Our beautiful 13	3184016	t	f	1	Yarra	-37.80086	144.97591	Entire home	Entire home/apt	8	2.5 baths	3
1049561370677	Flora - The three	Well presented,	760849	t	f	47	Bayside	-37.94615	145.01856	Entire home	Entire home/apt	6	2 baths	3
1281593036961	Peninsula family	Welcome to you	10483018	t	f	1	Frankston	-38.17300828	145.142421	Entire home	Entire home/apt	7	2 baths	4
1475065812322	Ivanhoe - luxury	Welcome to our	698288218	t	f	40	Banyule	-37.75359	145.05105	Entire home	Entire home/apt	6	2.5 baths	3
1583188816007	Cozy private ho	Kick back and re	193002320	t	f	1	Whittlesea	-37.54431	145.0139	Entire home	Entire home/apt	4	1 bath	3
1419210402669	Under the Vale	A quiet no-thru r	52283250	t	f	1	Port Phillip	-37.86529	144.98333	Entire home	Entire home/apt	4	1 bath	2
25592610	Beautiful & Mod	Massive 4 level	81969079	t	f	1	Yarra	-37.80253	144.99457	Entire home	Entire home/apt	5	2.5 baths	3
1373436978099	Daisy's Home	The house is loc	683388808	t	f	1	Bayside	-37.9057	144.99419	Entire home	Entire home/apt	6	2 baths	3
1419191572091	The Furniture Fe	A CREATIVE HA	26716551	t	t	21	Yarra	-37.7818838	144.9678737	Entire home	Entire home/apt	6	3 baths	3
14916459	60Luxury - 2mi	Overlooking Err	26093122	t	t	3	Melbourne	-37.80657	144.94981	Entire townhous	Entire home/apt	9	2.5 baths	4
48138050	Light-Filled Phot	The Studio is a s	46830585	t	t	1	Port Phillip	-37.83678	144.93337	Entire guesthou	Entire home/apt	6	1 bath	2
52713390	30%OFF5bedde	This outstanding	185076876	t	f	20	Knox	-37.88682	145.23763	Entire home	Entire home/apt	10	4.5 baths	5
1535267731925	Modern Treetop	Treetop Studio w	90541417	t	f	1	Manningham	-37.77270293	145.1005859	Entire home	Entire home/apt	5	1 bath	2
7405878404746	Warburton Stavs	Only a 2 minute	1801270	t	t	6	Yarra Ranges	-37.7559008	145.6814783	Entire home	Entire home/aot	8	2 baths	4

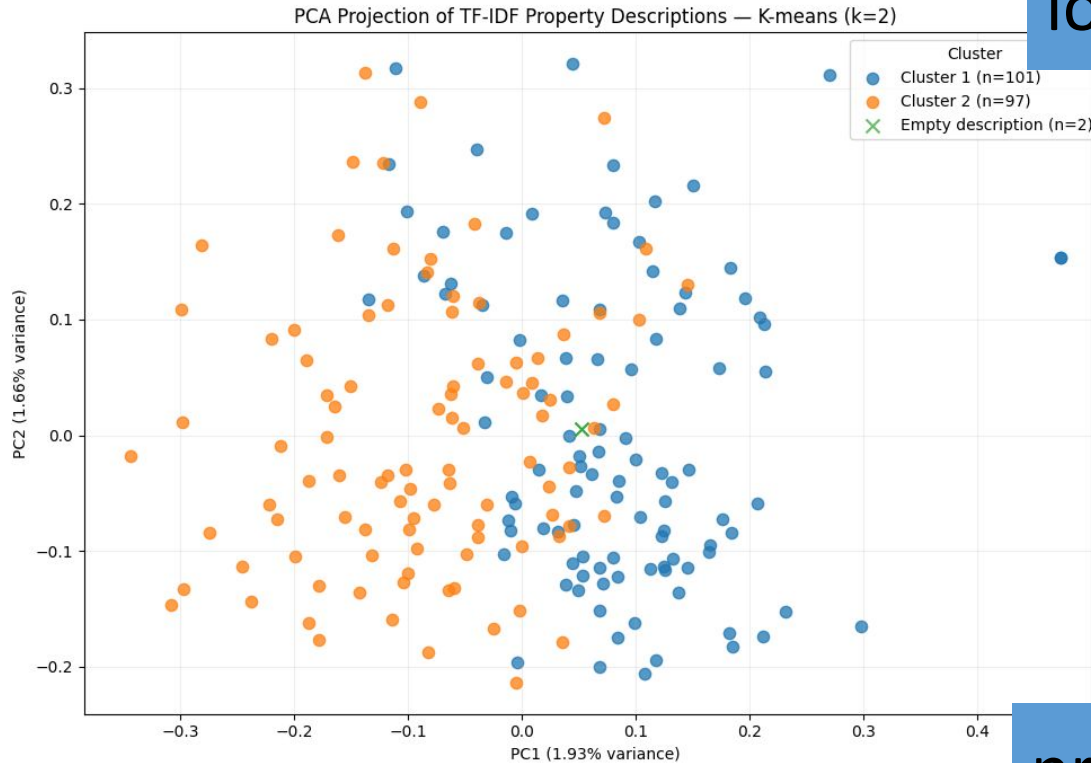
AirBnb Example



Cluster 1: minute,
away, city, walk,
drive, restaurant,
tram, train, yarra,
close, retreat

Cluster 2: bedroom,
bed, kitchen,
bathroom, living,
area, modern, open
plan, equip, tv,
parking, dining

AirBnb Example



Cluster 1: minute,
every city walk,
location features

tram, train, yarra,
close, retreat

Cluster 2: bedroom,
bed, kitchen,
bathroom, living,
area, modern, open
plan, equip, tv,

property features

Lecture outline

1. General introduction
2. k-Means clustering
3. **Visualisation of clustering tendency (VAT)**
4. Hierarchical clustering (Part 2)
5. Dimensionality reduction (Part 2)

Visual Assessment of Clustering Tendency (VAT)

- How many clusters does the data have naturally? How big are they? What is the likely membership of objects in each cluster?
 - The k parameter for k-Means
- Not always easy to know, especially if you have thousands of datapoints
- One approach: visually determine the clustering structure by inspecting a **heat map**

How do you create a **heat map**?

- Given n **data points** in m **clusters**, you must determine how **(dis)similar each one** is to the **cluster's centroid**.
- Display each cluster and each data point, and **colour the data point according to the level of (dis)similarity**
 - If two points are very similar → cool colour
 - If two points are very different → hot colour
- You must provide a *legend* (= table) that explains the amounts and their colours

What is a (dis)similarity matrix?

Object id	Feature1	Feature2	Feature3
1	5	10	15
2	10	5	10
3	20	20	20



- Use a vector distance function (e.g. Euclidean distance)
- Compute all pairwise distances between objects
- This gives a dissimilarity matrix

$$D(X, Y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + (x_3 - y_3)^2 + \dots (x_n - y_n)^2}$$

Object	1	2	3
1	0	8.7	18.7
2	8.7	0	20.6
3	18.7	20.6	0

$$\begin{aligned} & \sqrt{(5 - 10)^2 + (10 - 5)^2 + (15 - 10)^2} \\ &= \sqrt{25 + 25 + 25} = \sqrt{75} \approx 8.66 \rightarrow 8.7 \end{aligned}$$

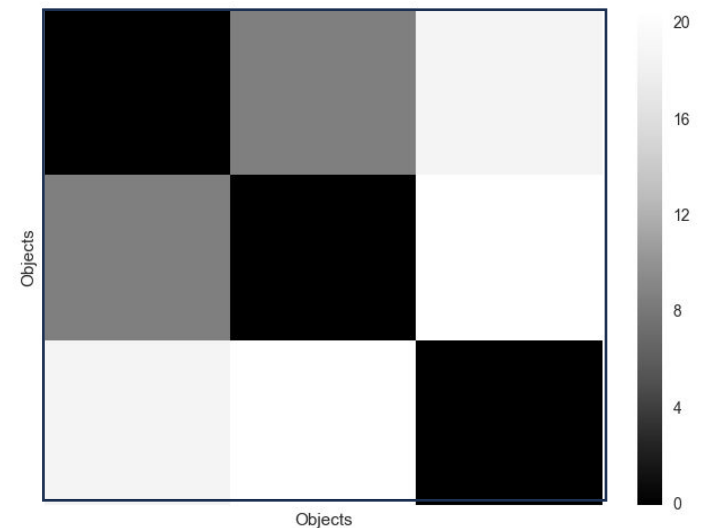
$$\begin{aligned} & \sqrt{(5 - 20)^2 + (10 - 20)^2 + (15 - 20)^2} \\ &= \sqrt{225 + 100 + 25} = \sqrt{350} \approx 18.7 \end{aligned}$$

$$\begin{aligned} & \sqrt{(10 - 20)^2 + (5 - 20)^2 + (10 - 20)^2} \\ &= \sqrt{100 + 225 + 100} = \sqrt{425} \approx 20.6 \end{aligned}$$

Visualising a dissimilarity matrix

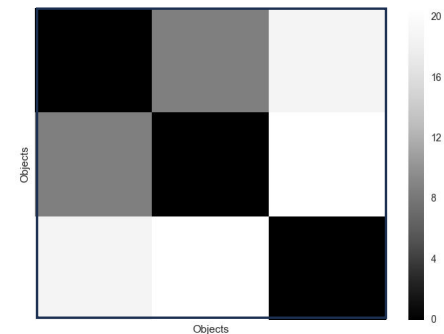
Visualise a dissimilarity matrix as a **heat map**, where the colour of each cell indicates that cell's value.

Object	1	2	3
1	0	8.7	18.7
2	8.7	0	20.6
3	18.7	20.6	0

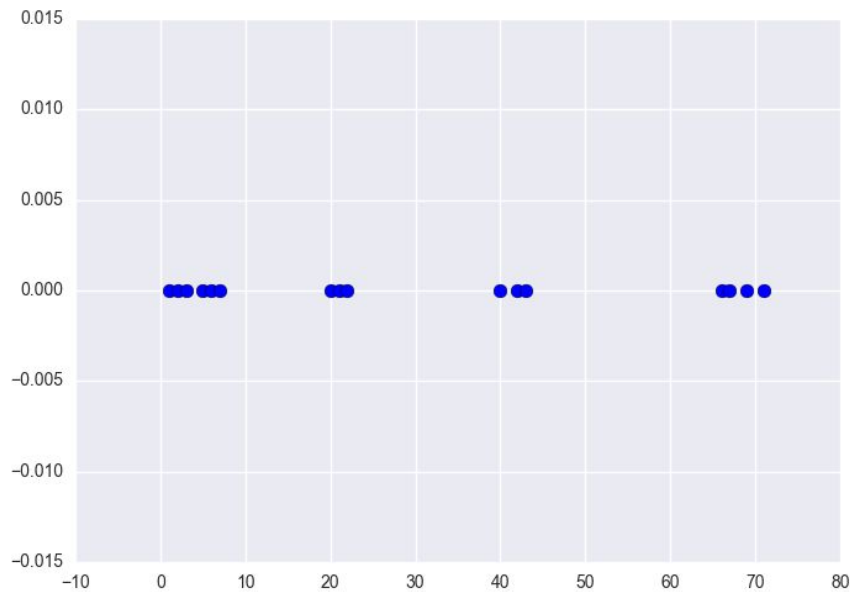


Properties of a dissimilarity matrix D

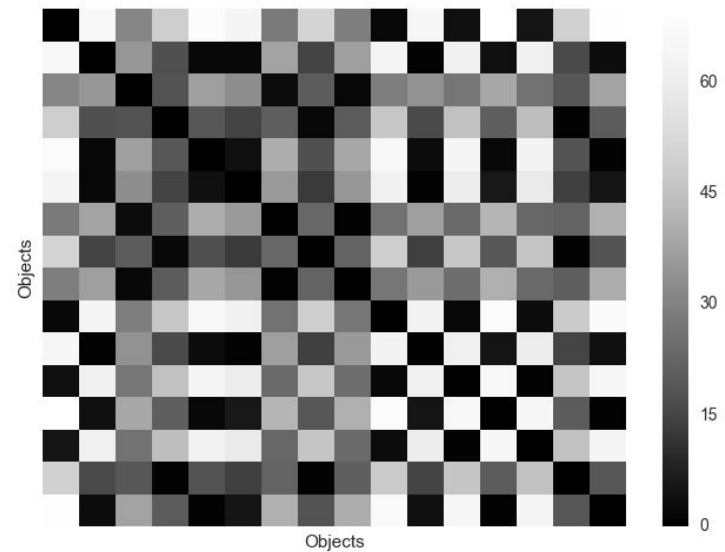
- The **diagonal of D** is all zeros
- D is symmetric about its leading diagonal
 - $D(i,j)=D(j,i)$ for all i and j
 - Objects follow the same order along rows and columns
- Visualising the (raw) dissimilarity matrix may **not reveal enough useful information**
 - **Further processing** often needed



Reordering a Dissimilarity matrix

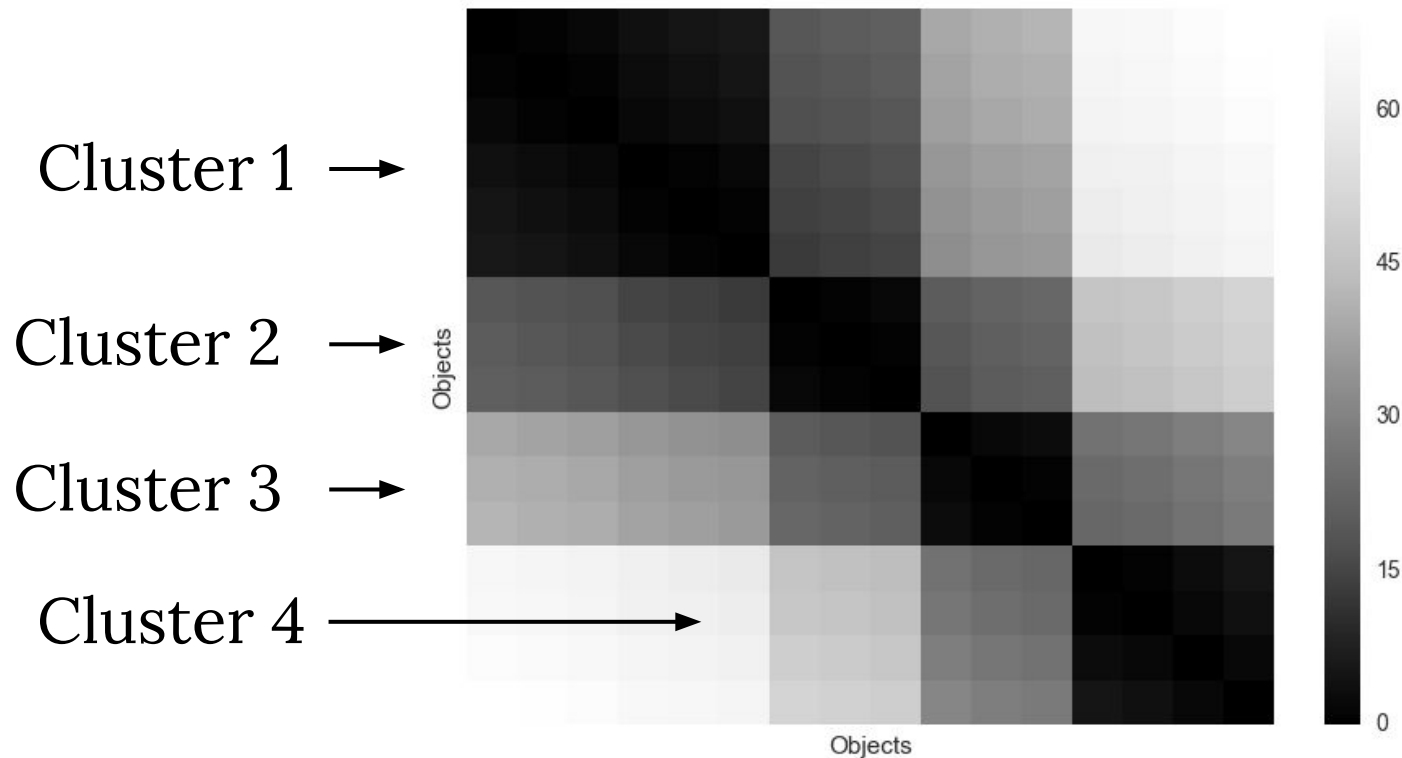


Example dataset
with 16 objects



Random order of the 16 objects
for the dissimilarity matrix

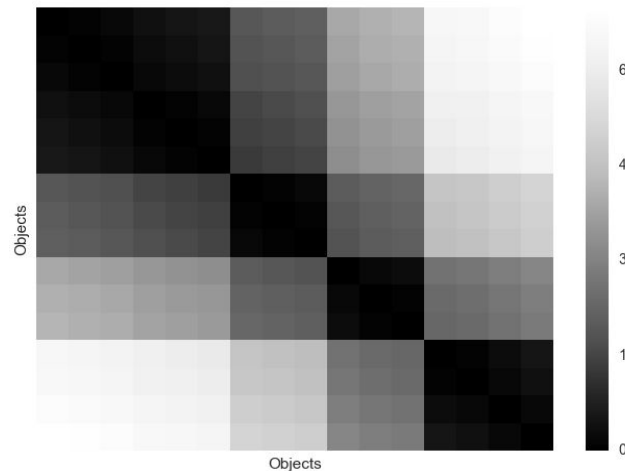
Reordering the matrix reveals clusters



A better ordering of the 16 objects. **Nearby objects in the ordering are similar to each other, producing large dark blocks.**

Good VAT images

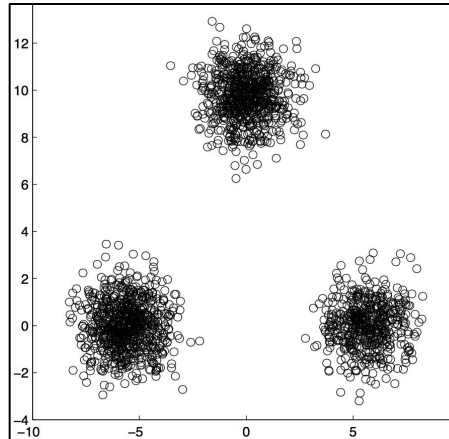
- A good VAT image suggests both the number of clusters and approximate members of each cluster
- A diagonal dark block appears in the VAT image only when a tight group exists in the data (= **low within-cluster dissimilarities**)



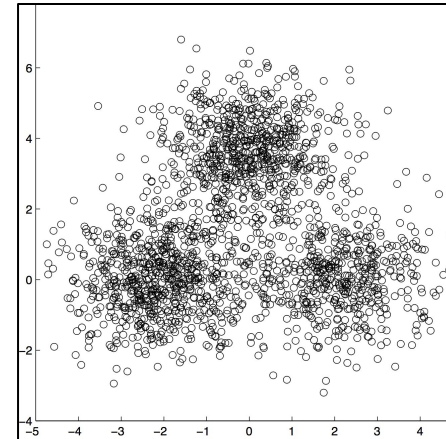
Exercise: Match the datasets with the VAT images

From: An algorithm for clustering Tendency Assessment, Hu, Y., Hathaway, R. WSEAS Transactions on Mathematics, 2008.

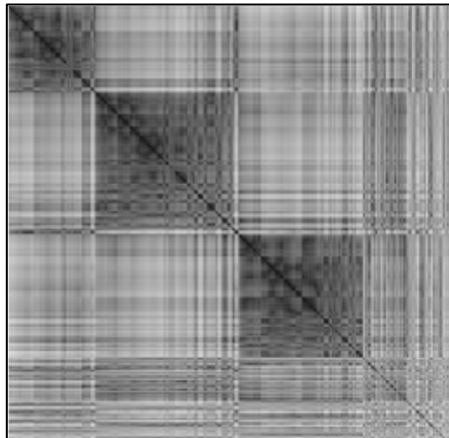
1



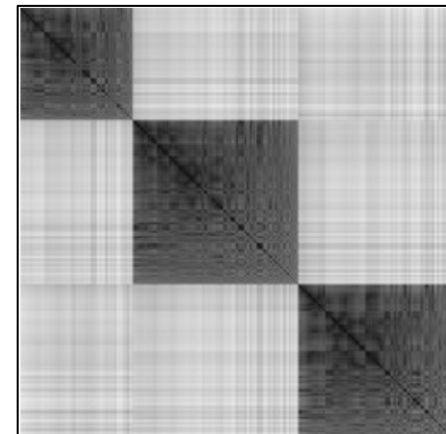
2



a



b



Obtaining best VAT order

- Goal: Reorder the data point pairs by similarity
- Visit each data point pair once
- Group together the closest ones first and then gradually include the ones further away
- How to do this?
 - **Start with the two most distant data point pairs**
 - Grow the group around one pair and leave all the still-untouched data point pairs with the other one
 - Continue until they are all seen
 - This reorders the data point pairs

VAT ordering example



VAT ordering example



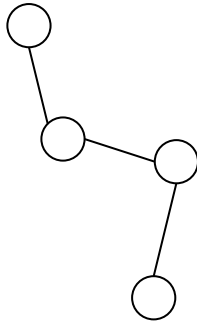
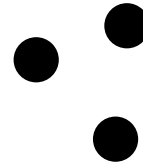
VAT ordering example



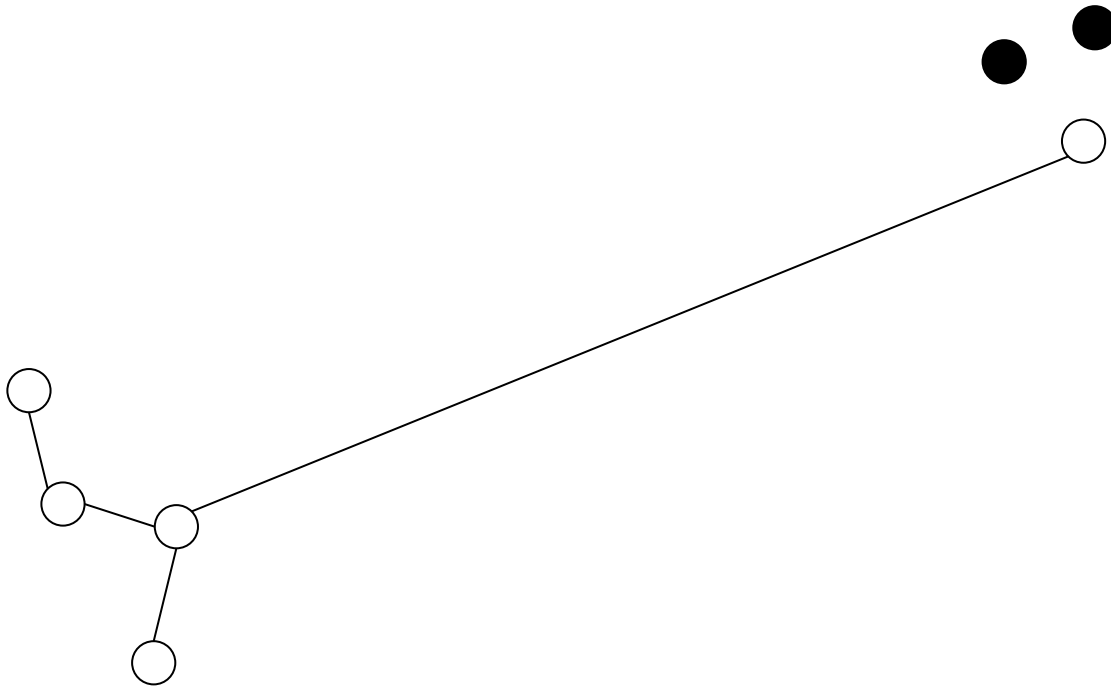
VAT ordering example



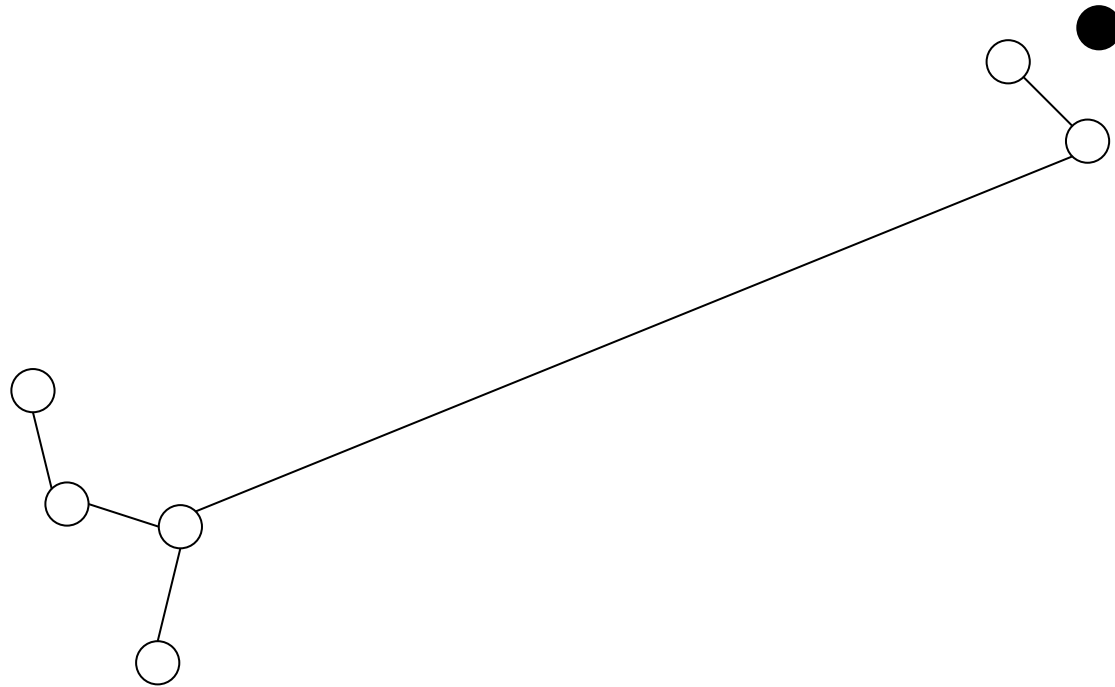
VAT ordering example



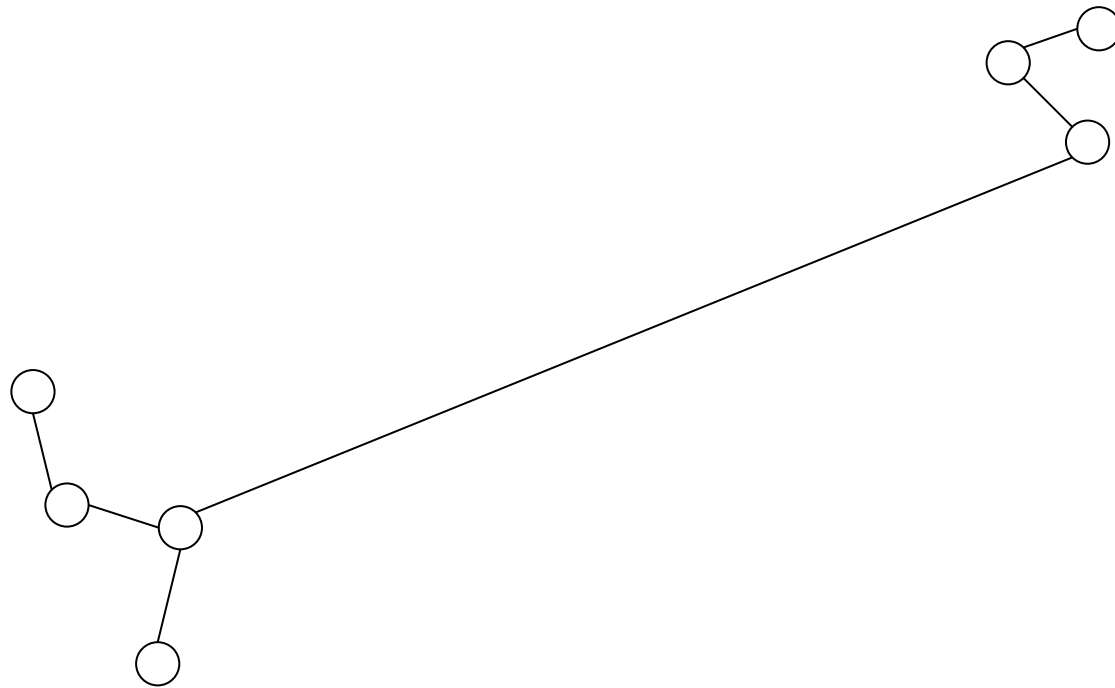
VAT ordering example



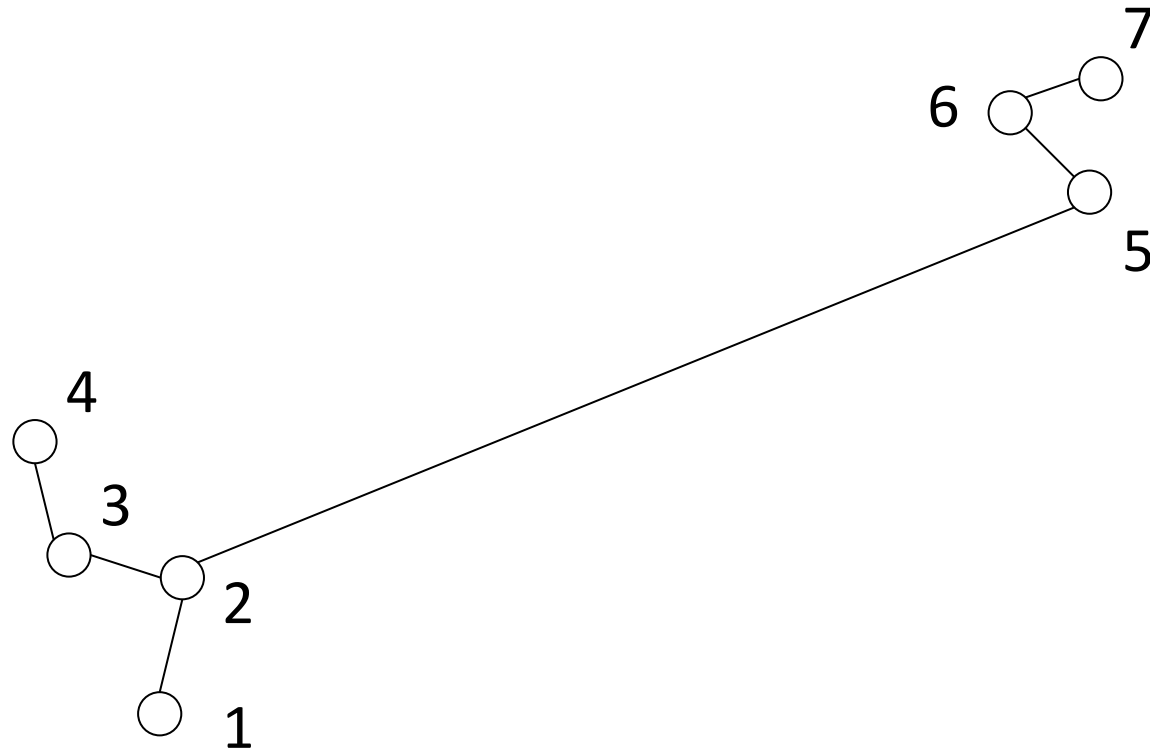
VAT ordering example



VAT ordering example



VAT ordering example



The VAT algorithm

(Bezdek and Hathaway 2002)

Given an $N \times N$ dissimilarity matrix D of data point pairs

Let $K=\{1, \dots, N\}$, $I=J=\{\}$ ### $\{\}$ is a set (collection of objects)

Pick the **two least similar objects** o_a and o_b from D

$P(1)=a$; $I=\{a\}$; $J=K-\{a\}$

For $r = 2, \dots, N$

Select (i,j) : pair of **most** similar objects o_i and o_j from D

Such that $i \in I, j \in J$ ### ' $\in$ ' means 'member of'

$P(r) = j$; $I = I \cup \{j\}$; $J = J - \{j\}$; ### ' $\cup$ ' means 'union' of two collections together

Obtain reordered dissimilarity matrix D^* from permutation (ordering) P

e.g. $P(1)=2, P(2)=5, P(3)=7$

The first object in the ordering is 2, the second is 5, the third is 7

VAT algorithm python code – you will practice in workshops

The VAT algorithm: problems

- VAT algorithm is not effective in every situation
- For complex-shaped datasets (either significant overlap or irregular geometries between different clusters), the quality of the VAT image may significantly degrade

